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Imperial College London

Department of Electrical and Electronic Engineering

EEELunarRover Report

ELEC40006 Electronics Design Project

Group 14

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Abstract

EEELunarRover is a remotely controlled rover designed to classify simulated lunar rocks by combining four interacting subsystems: an 89 kHz ASK radio receiver for age decoding, infrared event-rate sensing, 40 kHz ultrasonic detection, and magnetic-polarity sensing. The final architecture prioritised reliable subsystem integration: each analogue front end produces a microcontroller-compatible signal, the firmware fuses the sensor outputs into a rock-type decision, and the mecanum-wheel chassis provides short-range alignment without large turning corrections. Design decisions were evaluated against derived requirements for classification accuracy, manoeuvrability, mass, budget, electrical safety, and integration risk. The report also documents non-ideal outcomes, including infrared redesign, PCB-order delay, motor-shield supply failure, and the fallback stripboard/driver approach used to protect the final integration schedule.

Type	**Infrared rate**	**Ultrasound**	**Magnetic polarity**
Basaltoid	$\lambda = 547\ \mathrm{s}^{-1}$	40 kHz	Down
Gravion	$\lambda = 312\ \mathrm{s}^{-1}$	–	Down
Regolix	$\lambda = 312\ \mathrm{s}^{-1}$	40 kHz	Up
Lunarite	$\lambda = 547\ \mathrm{s}^{-1}$	–	Up

Table 1: Rock classification matrix used by the final multi-sensor fusion logic.

1 Project Management

1.1 Team Responsibilities

The project was decomposed into sensing, hardware, software, and integration workstreams because the principal risk was not an isolated circuit failure but a late mismatch between sensors, chassis, firmware, and user interface. Each subsystem therefore had an owner, while interface-critical work was deliberately shared so that mechanical dimensions, pin allocation, voltage levels, and data formats were agreed before final assembly.

Member	Primary role	Main responsibilities
Fred Ayres	Electrical engineer	Magnetic sensing, infrared design/testing, stripboard construction, power system support.
Owais Darwesh	Electrical engineer, editor	Radio receiver design and report writing.
Sijie Fei	Software engineer	Firmware, communication, rover control, multi-sensor fusion, web dashboard.
Zeyu Liu	Electrical engineer	Ultrasonic sensing, PCB design, construction and testing.
Colin Qian	Team leader, mechanical/electrical engineer, editor	3D design, assembly, PCB/IR/power support, control integration, project management and report editing.
Ximan Wang	Electrical engineer, editor	Infrared sensing design and presentation preparation.
All members	Integration and testing	System debugging, subsystem handover and final validation.

Table 2: Team roles and responsibilities.

1.2 Management Methods

Weekly planning meetings, a shared Notion workspace, WhatsApp for urgent coordination, and a budget/mass spreadsheet were used to maintain progress. Notion held the Gantt chart, design notes, task ownership and dependencies; WhatsApp was useful for fast debugging but was less reliable for traceable decisions, so important actions were moved back into Notion after meetings. The spreadsheet gave quantitative control of the two strongest project constraints: the 750 g mass limit and the £60 budget.

Constraint or dependency	Management response	Evidence / outcome
Mass below 750 g	Track major components before final assembly.	Planning subtotal was 489.9 g, leaving margin for chassis, wiring and fasteners.
Budget limit	Check orders and prioritise spares.	Approximately £47.82 was committed during planning; spare op-amps/comparators reduced repair risk.
Sensor-dependent chassis	Delay final mechanical layout until sensor geometry and circuit dimensions were known.	Initial chassis was completed on 13 May; final layout was frozen on 5 June after electronics updates.
PCB lead time	Avoid ordering before the infrared fault was understood.	PCB order was paused, revised on 6–7 June and placed on 8 June.
Motor-shield delivery risk	Prepare a fallback driver route.	The fallback TC78H620FNG H-bridge design became essential when the shield delivery moved to 28 June.

Table 3: Traceability between project constraints and management decisions.

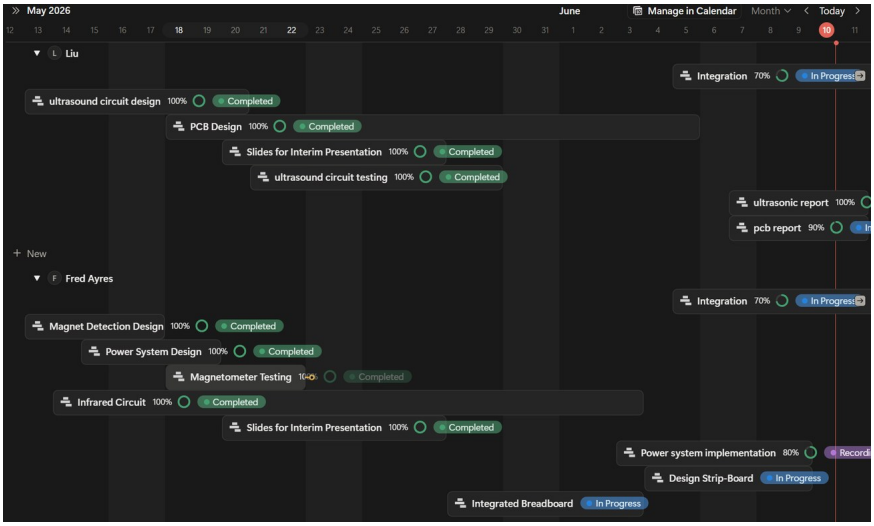


Figure 1: Notion Gantt chart used to track task progress, deadlines and subsystem dependencies.

Total spent	47.82		Planned spending	0.00							
Total weight	489.9										
Group	Name	price	weight/g	total weight	Count	Source	Status	Order from	Order code	Spent	Note
Hardware	Yellow motor	0.00	30	119.8	4	eeebug	Using				
	Mecanum wheel	0.00	50	197.2	4	3d print	Using				60mm, pla
	Rivet	3.95	-	-	50	Rapid(shop)	Using	Rapid	86-1564	3.95	shaft
	Chassis	0.00	-	-	1	3d print/laser cut	Using				pla
	adafruit & shield	0.00	94.3	39.3	1	extension kit	Using				
	battery	0.00	26.1	104.4	4	eeebug	Using				
	battery holder	0.00	14.6	14.6	1	eeebug/3d print	Using				
	pcb	5.00	-	-	1	shop	Using			0	independent budget
	Sensors	0.00	14.6	14.6	1	Circuit	Using				
Circuit	SL0190 -- CA3140	2.72			1	shop	Purchased	eedstores	SL0190	2.72	opamp
	SI0160 -- LM311N	0.54			1	shop	Purchased	eedstores	SI0160	0.54	opamp
	MCUSD16A40S12RO	4.31			1	external	Using	onecall	SN36976	1.78	ultrasonic transducer
	SEN4019(*SEN0419)	6.07			1	Farnell(shop)	Using	onecall	4308272	5.06	3 axis magnetometer
	WPSH207	11.63			1	Farnell(shop)	Discard	Conrad	2481967 - 62	11.63	Motor Driver Shield
	MCP6022 I/P x5	8.40			1	RS components	Using	RS	403-064	8.4	opamp
	SFH 314 FA	3.43			10	Onecall(shop)	Using	onecall	2981735	3.43	infrared sensor
	MCP6022 I/P x3	1.47			3	Onecall(shop)	Using	onecall	9758640	4.41	opamp
	Stripboard	9.11			1	RS components	Using	RS	100-4328	9.11	stripboard
	MCP6022 I/P x3	1.47			3	Onecall(shop)	Purchased	onecall	9758640	4.41	opamp
	MCP6541	4.01			1	RS components	Purchased	RS	402-790	4.01	comparator

Figure 2: Budget and mass spreadsheet used to monitor cost and weight during design.

1.3 Delays and Risk Management

The main delays were caused by dependency order: the chassis needed final sensor geometry, the PCB needed a working infrared circuit, and the drivetrain needed a motor driver that actually arrived before integration. The team responded by delaying irreversible decisions when the risk of a wrong build exceeded the schedule cost. For example, ordering the PCB before resolving the infrared fault would have locked an unreliable front end into the manufactured board; similarly, waiting for the delayed WPSH207 motor shield would have prevented drivetrain testing.

Risk	Impact	Mitigation	Evaluation
Infrared circuit failed during testing	PCB could have captured an incorrect analogue design.	Pause PCB order, redesign and retest the IR front end.	Delayed fabrication but avoided a likely board-level failure.
Motor shield delayed	Rover drive system could not rely on the planned shield.	Use available dual H-bridge driver ICs.	Preserved movement and PWM testing before the demo.
Mechanical layout frozen too early	Sensors or boards might not fit.	Keep CAD parametric and finalise after electronics dimensions were known.	Improved packaging but compressed assembly time.
Informal task communication	Actions could be lost in message history.	Move decisions and owners into Notion after discussion.	Improved traceability for later integration.

Table 4: Risk register entries with mitigation and outcome.

1.4 Backup Plans

The backup strategy was to preserve a demonstrable integrated rover even if the most polished implementation route failed. The motor-control fallback replaced the unavailable shield with TC78H620FNG H-bridge drivers. The sensor fallback was to use the strongest reliable subset of rock-type evidence while clearly displaying confidence and raw readings to the operator. The construction fallback retained stripboard for high-risk or heat-sensitive circuitry while the PCB provided a cleaner final integration route for the main analogue stages. These decisions were less elegant than the original plan but were appropriate engineering trade-offs because they protected system-level functionality, which is the dominant criterion in the final demonstration.

2 Sensors

2.1 Radio Detection

The rock transmits its age using a radio signal with an 89 kHz carrier frequency. The signal uses two-level amplitude-shift keying, meaning the carrier is switched on and off to represent digital 1s and 0s. The data is sent at 600 bits per second as ASCII characters inside UART packets, with each character framed by 1 start bit and 1 stop bit. Each age message is four characters long and begins with #; for example, #123 represents an age of 1.23 billion years.

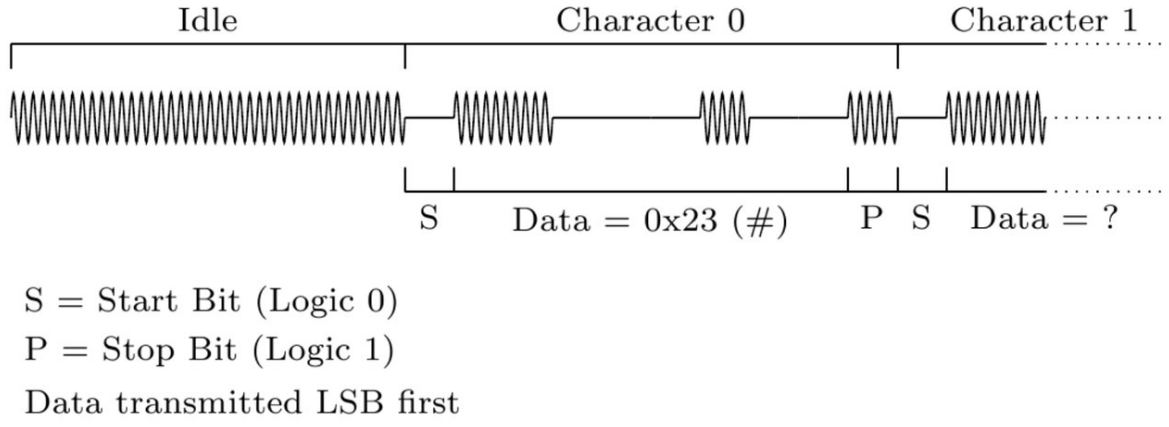


Figure 3: Radio Signal Waveform

To determine the specific age of each rock, the rover must intercept and decode a telemetry signal transmitted at a carrier frequency of 89 kHz, modulated using two-level amplitude-shift keying (ASK). This age metric is a core deliverable, requiring the software team to parse the decoded ASCII characters into a UART packet. To ensure robust signal detection, the receiving subsystem must address the following technical challenges:

- Filtering: Attenuate environmental noise by resonating strictly at the 89 kHz target frequency.
- Signal Amplification: Amplify low-amplitude induced signals to a usable voltage range while maintaining a high signal-to-noise ratio (SNR).
- Signal Conditioning: Rectify, smooth, and digitise the analogue waveform so the microcontroller can reliably process the 600 bit/s UART stream.



Figure 4: High-level block diagram illustrating the sequential signal processing stages of the radio detection architecture, from the initial coil antenna interception to the final digital UART input.

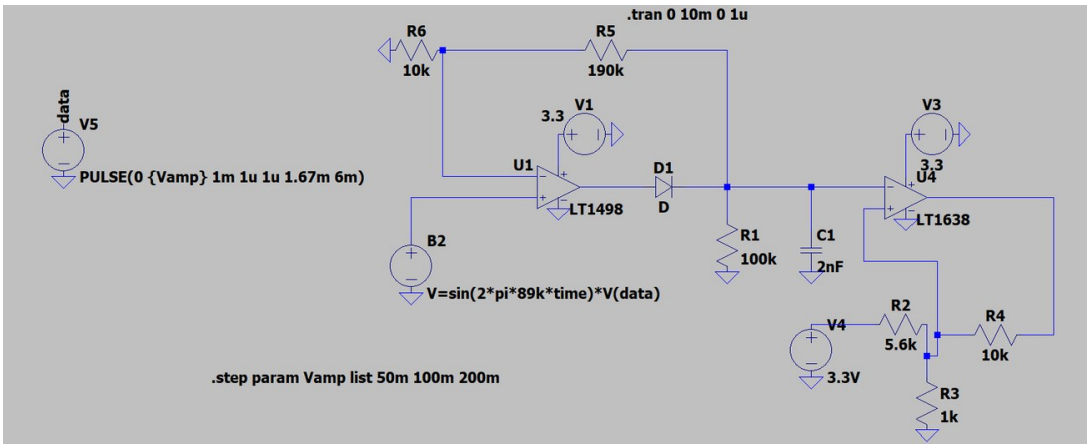


Figure 5: Complete schematic of the radio detection subsystem, detailing the active bandpass filtering, envelope detection, and Schmitt trigger digitisation stages.

2.1.1 Antenna Design

The primary parameters considered during the antenna design phase were the loop diameter and the total number of turns. Analytically, the relationship between these physical dimensions and the induced electromotive force (EMF) is governed by Faraday's Law of Induction:

$$V = -N\frac{d\phi}{dt}$$

Therefore, a larger coil area or a greater number of turns yields a higher induced voltage, significantly simplifying signal detection. However, increasing either variable introduces a direct trade-off by adding physical mass. To ensure the rover remains strictly under the 750g weight limit required to safely traverse the lunar surface weight-sensitive zone, a smaller, lightweight coil diameter was ultimately selected. To compensate for the reduced signal amplitude, active amplification stages were integrated into the circuit design, and the rover's navigation algorithm will be tasked with manoeuvring physically closer to the signal source.

2.1.2 Tuning Capacitor

To isolate the target 89 kHz carrier wave, an air-core inductor was custom-wound and its inductance was empirically measured. To achieve maximum voltage sensitivity at this specific frequency, the inductor was paired with a tuning capacitor to create a parallel LC resonant tank circuit. The required capacitance was calculated by rearranging the standard resonant frequency formula:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \Rightarrow C = \frac{1}{4\pi^2 f_0^2 L}$$

Given the measured coil inductance (L) of 96 µH and the target resonant frequency (f0) of 89 kHz, the ideal capacitance was calculated to be approximately 33.3 nF. To match this requirement as closely as possible using standard off-the-shelf component values, a 33 nF capacitor was selected for the physical implementation.

2.1.3 Rectification

Following frequency isolation, the AC signal must be rectified. Several rectification topologies were evaluated against the constraints of varying operational distances and the low-amplitude nature of the induced signal. The findings are summarised in Table 5.

Topology	**Minimum useful input**	**Decision**
Standard half-wave diode	About 0.7 V	Rejected because the forward drop would remove weak received signals.
Schottky rectifier	About 0.2 V	Rejected because improved sensitivity was still not sufficient at the target range.
Precision active rectifier	Near zero ideal threshold	Selected because the op-amp stage removes the diode-drop limitation and can provide gain.

Table 5: Rectifier topology decision matrix for the radio receiver.

2.1.4 Envelope Detection

Once rectified, the 89 kHz carrier wave must be stripped away to reveal the underlying 1-bit ASK envelope. A passive RC envelope detector was implemented, with the resistor and capacitor values selected specifically to balance the carrier period and the bit transmission rate.

- The frequency of the carrier wave is 89 kHz, resulting in a period (T_c) of 11.2 µs.
- The data transmission rate is 600 bps, resulting in a bit period (T_{bit}) of 1.67 ms.

To satisfy the operational condition $T_c \ll \tau \ll T_{bit}$, component values of R = 100 kΩ and C = 3 nF were chosen, yielding an RC time constant (τ) of 300 µs. Because 300 µs is significantly larger than 11.2 µs, the capacitor discharges slowly enough to hold the signal HIGH between high-frequency peaks, effectively removing the ripple. Conversely, because 300 µs is much smaller than the 1.67 ms bit period, the capacitor has ample time to drain fully to 0 V when the rock stops transmitting a "1". This prevents the voltage from bleeding into the next "0" bit.

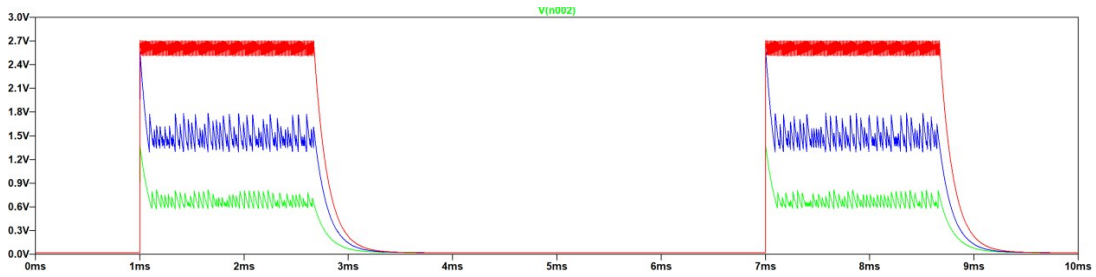


Figure 6: LTspice transient simulation at Point A. The multiple traces demonstrate the circuit's envelope detection response to an 89 kHz carrier wave stepped at varying input magnitudes (50 mV, 100 mV, and 200 mV), successfully smoothing the high-frequency signals into distinct DC pulses representing a logical '1'.

2.1.5 Schmitt Trigger

While the envelope detector successfully smooths the signal, the resulting waveform contains finite transition times and residual analogue ripple. To ensure reliable UART decoding by the software team, this signal must be strictly digitised. A Schmitt trigger was implemented to convert the analogue envelope into a crisp, logic-level square wave.

This design choice was highly cost and space-efficient, as it utilised the remaining free operational amplifier within the dual op-amp package.

The circuit uses an inverting topology, operating on a 3.3V supply. By utilising a positive feedback resistor network (comprising 27.5 kΩ, 349 Ω, and a 10 kΩ feedback resistor), the circuit establishes two distinct switching thresholds to provide noise immunity:

- Lower threshold: approximately 40 mV.
- Upper threshold: approximately 150 mV.

Finally, a 100 nF decoupling capacitor was placed in parallel with the reference voltage divider. This ensures that high-frequency noise from the power supply does not destabilise the precisely calculated switching thresholds.

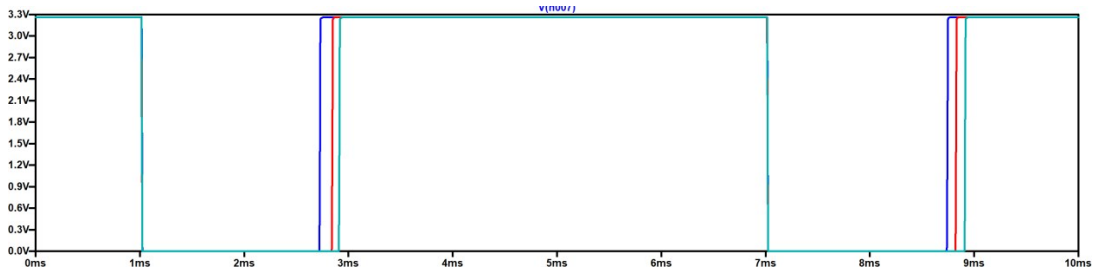


Figure 7: LTspice transient simulation at Point B. The output demonstrates the Schmitt trigger's response to the envelope signals derived from varying input magnitudes (50 mV, 100 mV, and 200 mV). The traces show a clean, robust 3.3V digital square wave, completely free of analogue ripple, ensuring reliable logic-level transitions for the MCU's UART serial pins.

2.2 Ultrasonic Detection

The ultrasonic subsystem converts the presence of a 40 kHz signal emitted by a rock simulator into a single 3.3 V digital classification bit for the rover controller. The project guide states that rock type can be indicated by ultrasonic signal presence or absence at 40 kHz; it also specifies that the Metro M0 logic pins must not exceed 3.3 V, with logic-low at 0 V and logic-high at 3.3 V. The design therefore prioritised three requirements: reliable sensitivity to an attenuated 40 kHz signal, rejection of noise and non-ultrasonic content, and a microcontroller-safe digital output.

2.2.1 Requirements and Design Process

The design was developed as a signal-conditioning chain rather than as a software-sampled ultrasonic receiver. This choice was made because the received carrier amplitude is small after passing through the rock shell, while the required output is only binary. The signal chain is:

1. **Transducer pickup:** an MCUSD16A40S12RO ultrasonic transducer receives the rock signal at its 40 kHz resonance.
2. **AC coupling:** a 100 nF capacitor blocks any DC offset produced by the transducer before amplification.
3. **Band-pass gain:** an active band-pass stage centred at 40 kHz amplifies the desired carrier while attenuating low-frequency and broadband noise.
4. **Additional gain:** a second inverting gain stage increases sensitivity to the millivolt-level received signal.
5. **Envelope detection:** a passive rectifier and RC network converts the 40 kHz carrier into a slowly varying voltage proportional to received amplitude.
6. **Comparator:** a low-threshold comparator converts the envelope into a clean digital output: HIGH for ultrasonic signal present and LOW for absent.

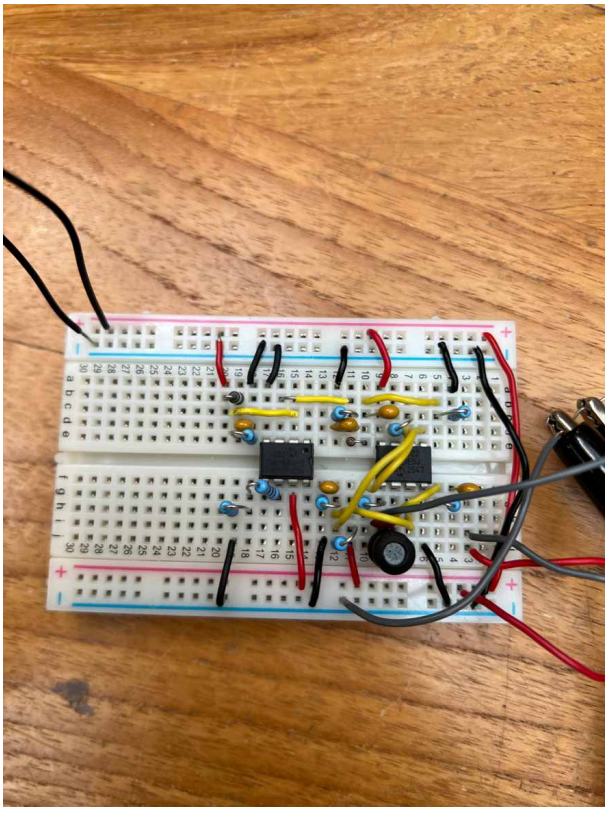


Figure 8: Breadboard Prototype

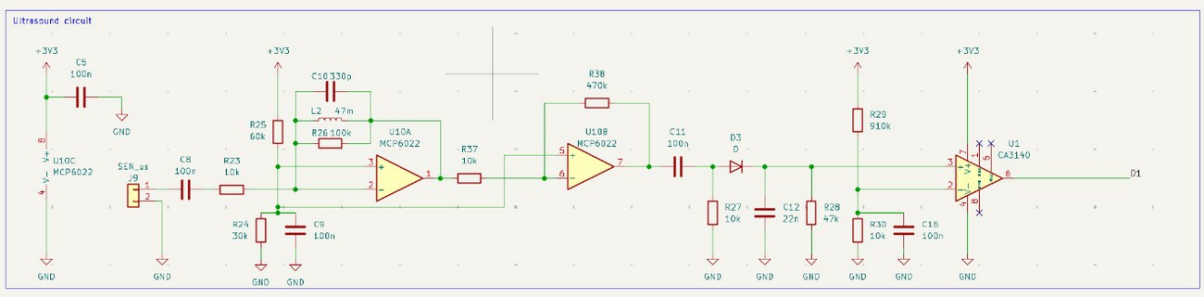


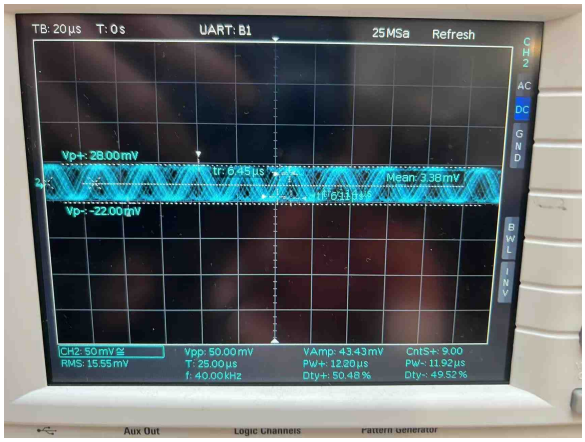
Figure 9: KiCad signal-chain schematic

The initial mechanical concept was to mount the ultrasonic transducer on a servo and actively aim it towards the rock drill hole. Testing showed that this would add mechanical complexity without being necessary: adding a further electronic gain stage allowed the transducer to detect the signal when placed near and facing the rock. This was a better design trade-off for an integrated rover because it reduced moving parts, control complexity, and alignment risk, while still providing the binary information required for rock classification.

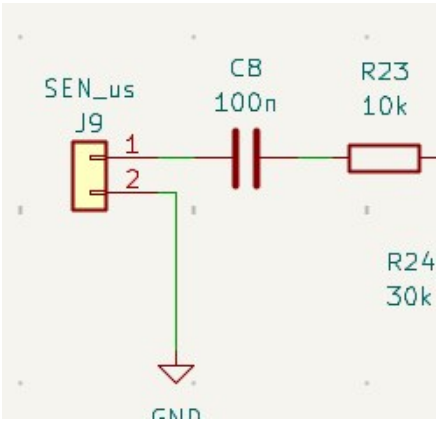
The supply strategy was also revised during prototyping. A 5 V supply gave stable readings up to roughly 0.5 m, but this range was excessive for the final arena because nearby rocks could produce false detections. The subsystem was therefore moved to 3.3 V operation. This made the output directly compatible with the Metro M0, reduced level-shifting requirements, and intentionally biased the design towards local proximity detection rather than long-range detection.

2.2.2 Implementation

Input coupling and filtering. Without the rock shell, the received signal was measured at about 300 $mV_{p\text{-}p}$. With the shell present, the signal was attenuated by approximately a factor of 10 to around 50 mV. The input therefore required high gain, but not broadband gain: amplifying all frequencies would increase susceptibility to electrical noise and other signals in the rover environment.



(a) Transducer signal measurement.



(b) 100 nF AC-coupling capacitor.

Figure 10: Front-end evidence: the transducer output is small enough after shell attenuation to require active gain and filtering.

An inverting active band-pass topology was selected using the MCP6022 dual op-amp. Its peak gain is set approximately by the feedback-to-input resistance ratio, but the feedback resistance also changes bandwidth and quality factor. The band-pass stage gain was therefore

limited so that the filter retained a practical bandwidth of approximately 5 kHz and a quality factor near 8 around the 40 kHz carrier. The finite impedance of the practical LC network at resonance, measured at roughly 100 kΩ, also constrained the design: the input impedance needed to remain at least around 10 kΩ to avoid loading the transducer, so the first-stage gain was kept below about 10.

A separate inverting amplifier was added after the band-pass stage. Combined with the band-pass stage, the implemented gain was approximately 470. Because the subsystem only detects whether the carrier is present, waveform distortion at high signal levels was acceptable; the important metric was reliable separation between noise and a true ultrasonic signal. Both gain stages were biased at approximately 1.1 V, below the 3.3 V midpoint, giving more positive headroom for the envelope detector that follows.

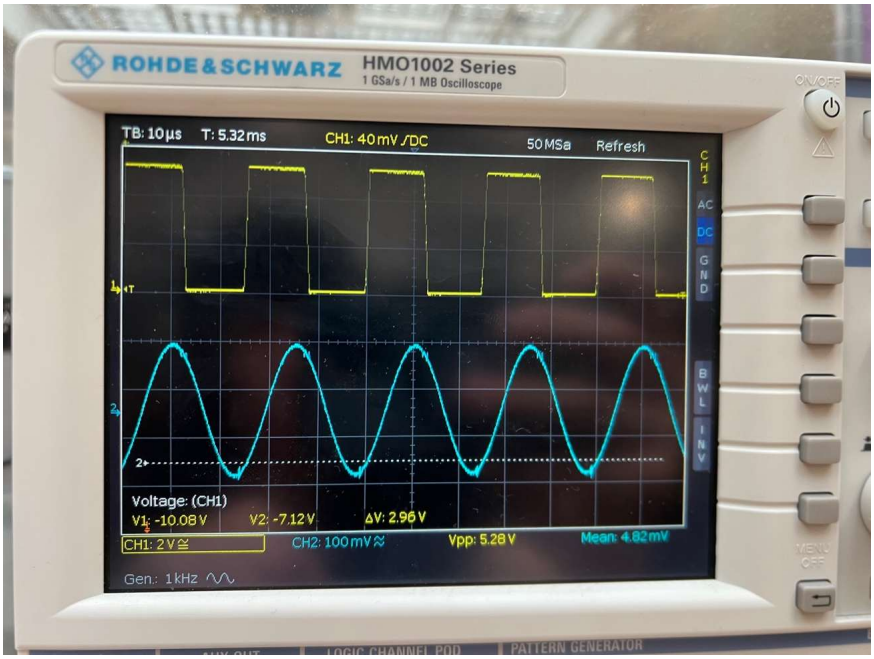
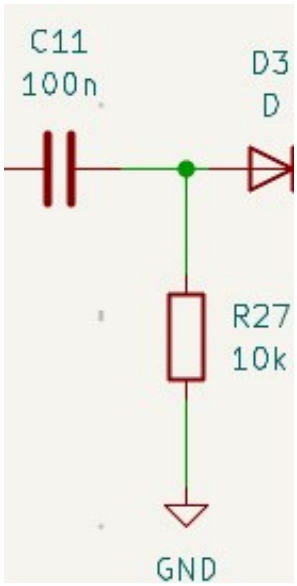


Figure 11: Measured gain-stage behaviour. The overall gain was documented as approximately 470, sufficient to lift the shell-attenuated millivolt input into a detectable range.

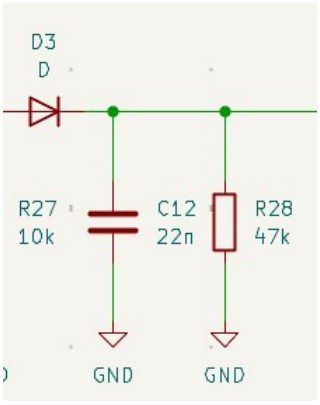
Envelope detector. After amplification, the circuit removes the op-amp bias using a high-pass filter with a cutoff near 159 Hz. The resulting AC signal is rectified and smoothed by a passive RC envelope detector. The documented design used a 22 nF capacitor and a 10 kΩ discharge path, giving

$$\tau = RC = 10\,\text{k}\Omega \times 22\,\text{nF} = 220\,\mu\text{s}.$$

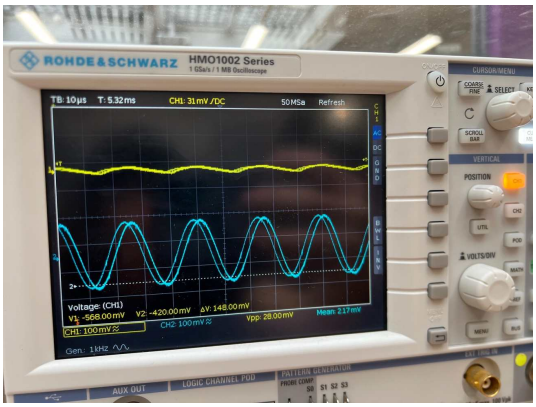
At 40 kHz, the carrier period is 25 µs, so this time constant is about 8.8 carrier periods. It is therefore long enough to smooth the carrier ripple while remaining short enough for the digital output to respond on a sub-millisecond scale. The schematic also includes a 47 kΩ load/discharge resistor, so the exact decay rate depends on the final assembled network; the key design condition is that the envelope time constant is much larger than the ultrasonic period.



(a) Bias-removal high-pass filter.



(b) Envelope detector.



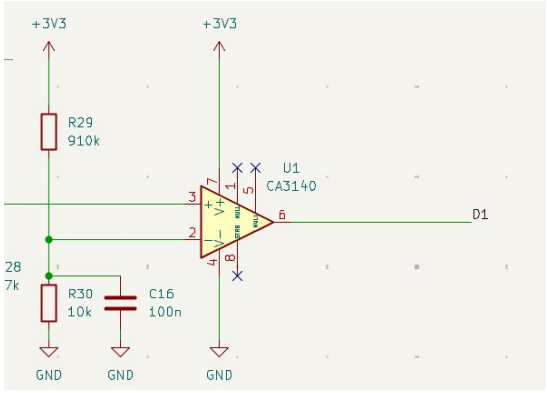
(c) Envelope output.

Figure 12: Envelope detector implementation. The oscilloscope evidence shows the carrier converted into a near-DC amplitude signal, with the expected diode voltage drop.

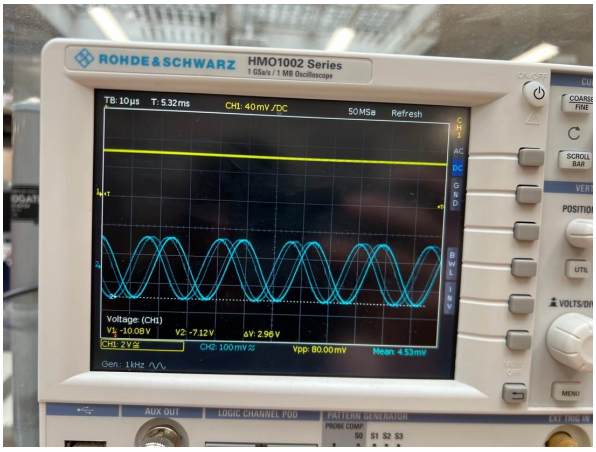
Comparator and digital interface. The envelope output was reported to range from about 100 mV to above 1 V when the ultrasonic signal was present. A threshold of approximately 36 mV was generated with a high-ratio divider from the 3.3 V rail:

$$V_{\text{th}} = 3.3\,\text{V}\frac{10\,\text{k}\Omega}{910\,\text{k}\Omega + 10\,\text{k}\Omega} \approx 35.9\,\text{mV}.$$

This threshold sits above the observed noise floor while remaining comfortably below the envelope amplitude produced by a true received signal. The comparator was implemented with an MCP6541 in the physical circuit; the KiCad schematic uses a CA3140 symbol because it was available in the library and shares the required pinout. The comparator output is interpreted directly by the microcontroller: HIGH at 3.3 V when the 40 kHz carrier is present, LOW at 0 V otherwise.



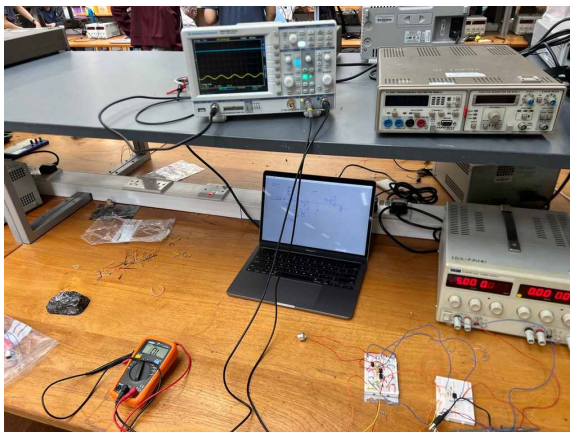
(a) Comparator threshold network and symbol.



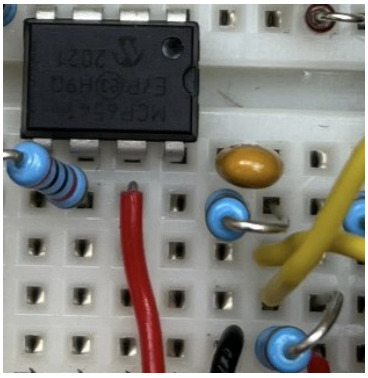
(b) Comparator output: HIGH when ultrasonic signal is present.

Figure 13: Comparator stage converting the analogue envelope to a Metro-compatible digital rock-type bit.

Decoupling capacitors were added to the power rails and bias nodes to reduce supply noise and bias drift. This is important because the circuit contains high gain before thresholding: any supply-coupled noise at the op-amp inputs can be amplified and create false comparator transitions.



(a) Bench setup used during range and supply-voltage trials.



(b) Prototype decoupling.

Figure 14: Integration-focused prototype refinements: supply selection and decoupling improved compatibility with the rover electronics.

2.2.3 Evaluation

The subsystem was evaluated at each stage rather than only at the final digital output. This is appropriate because failures in the transducer, gain/filter, envelope detector, or comparator can produce the same final symptom: an incorrect binary rock-type bit. The evidence collected is summarised in Table 6.

Table 6: Requirement-level evaluation of the ultrasonic detection subsystem.

Requirement	Test or evidence	Measured/observed result	Outcome
Detect the 40 kHz ultrasonic carrier	Oscilloscope measurement at the transducer and after amplification	The unshielded signal was about 300 $mV_{p\text{-}p}$; with the rock shell it fell to about 3–5 mV, confirming the need for high gain.	Met
Reject unwanted frequencies and noise	Active band-pass design centred at the carrier frequency	Band-pass centred at 40 kHz, with approximate bandwidth 5 kHz and $Q \approx 8$. Gain of this stage was limited to preserve bandwidth and input impedance.	Met
Amplify the attenuated signal enough for thresholding	Cascaded active gain stages	Overall gain of approximately 470 produced envelope amplitudes from about 100 mV to above 1 V, well above the comparator threshold.	Met

Continued on next page

Requirement	Test or evidence	Measured/observed result	Outcome
Convert carrier amplitude to a slow analogue signal	Envelope detector oscilloscope trace	RC detector produced a near-DC envelope; the observed diode drop was approximately 0.7 V, consistent with a passive rectifier.	Met
Provide a clean microcontroller-safe digital bit	Comparator output test	Threshold was set to about 36 mV; output switched to 3.3 V when ultrasonic signal was present and 0 V when absent.	Met
Avoid false detections in dense rock placement	Supply-voltage and range trial	At 5 V, detection worked up to about 0.5 m; this was judged too sensitive for the demo environment, so the circuit was moved to 3.3 V to favour proximity detection and direct logic compatibility.	Improved; final arena test required
Maintain stable bias and supply rails	Prototype inspection and schematic update	Decoupling capacitors were added to rails and bias points to reduce false switching from supply or bias noise.	Met in prototype

The strongest result is the clear separation between the envelope amplitude when a 40 kHz signal is present and the 36 mV comparator threshold. Even the lower reported envelope value of approximately 100 mV gives a margin of about 64 mV, while stronger detections exceed 1 V. This margin explains why a single-threshold comparator was sufficient during bench tests and why the final output remained stable rather than flickering near the switching point.

The main limitation is that bench range is not the same as integrated rover performance. The 5 V version demonstrated sensitivity, but the later 3.3 V version was selected because excessive range could classify the wrong rock in a dense field. The final report should therefore include an integrated test in which the rover approaches multiple rocks at realistic spacing and records true positives, false positives, and missed detections against distance and angle. The most useful final metrics would be: maximum reliable detection distance, minimum spacing before cross-detection occurs, angular tolerance of the transducer, and digital-output stability over repeated trials.

Overall, the subsystem satisfies its core electrical requirements: it detects a weak 40 kHz ultrasonic signal, filters and amplifies it using a technically justified analogue signal chain, converts the carrier into an envelope, and outputs a safe 0 V/3.3 V binary signal for rock-type classification. The design process also shows meaningful iteration: the servo aiming mechanism was removed in favour of electronic gain, and the supply voltage was changed from 5 V to 3.3 V after evaluating the practical risk of false detections in the final demonstration environment.

2.3 Infrared Pulse Detection

2.3.1 Role and requirements

The infrared subsystem estimates the random optical-event rate emitted by the rock simulator. Basaltoid and Lunarite produce the high-rate class at approximately $\lambda = 547\ \mathrm{s}^{-1}$, while Gravion and Regolix produce the low-rate class at approximately $\lambda = 312\ \mathrm{s}^{-1}$. Infrared sensing is therefore a radioactivity-class discriminator rather than a complete rock identifier; the final rock type is obtained only after fusing it with the ultrasonic and magnetic decisions. The hardware

requirement was to detect weak near-infrared pulses of roughly 50 µs, reject slowly varying ambient light and mains-light ripple, and deliver a clean 0–3.3 V pulse train to a Metro M0 interrupt input.

2.3.2 Counting method

The useful measurement is pulse rate rather than pulse amplitude. For a fixed counting window T,

$$N(T) \sim \text{Poisson}(\lambda T), \qquad \text{E}[N] = \lambda T, \qquad \sigma_N = \sqrt{\lambda T}.$$

Using the midpoint rate $\lambda_{\text{th}} = (312 + 547)/2 = 429.5\ \text{s}^{-1}$, the firmware threshold is $N_{\text{th}} = \lceil 429.5T \rceil$. A 0.5 s window gives expected counts of 156 and 273.5, with a threshold of 215. This provides large statistical separation while keeping the operator delay short; longer windows remain a fallback if real hardware losses reduce the separation.

Table 7: Expected infrared-count separation for candidate software counting windows. Probabilities assume ideal pulse detection, so hardware noise and missed edges must still be validated experimentally.

Window	Low-rate mean	High-rate mean	Threshold	Worst ideal error
0.12 s	37.4	65.6	52	3.6 %
0.25 s	78.0	136.8	108	0.49 %
0.50 s	156.0	273.5	215	0.011 %

2.3.3 Analogue front-end design and iteration

The final signal chain was: phototransistor receiver, AC coupling, two MCP6022 gain stages, comparator thresholding and GPIO interrupt counting. The first prototype used an SFH 300 phototransistor, a high-gain non-inverting stage and an open-collector comparator pulled up to 3.3 V. It showed oscilloscope activity, but it was not reliable on the moving rover because ambient-light variation shifted the baseline, weak covered-rock pulses were easily slewed, and excessive single-stage gain made resistor thermal noise and amplifier noise significant. The redesigned circuit split the gain across two lower-resistance stages, inserted AC coupling between stages, added local decoupling and swapped the comparator polarity so that each detected event produced a positive-going digital pulse.

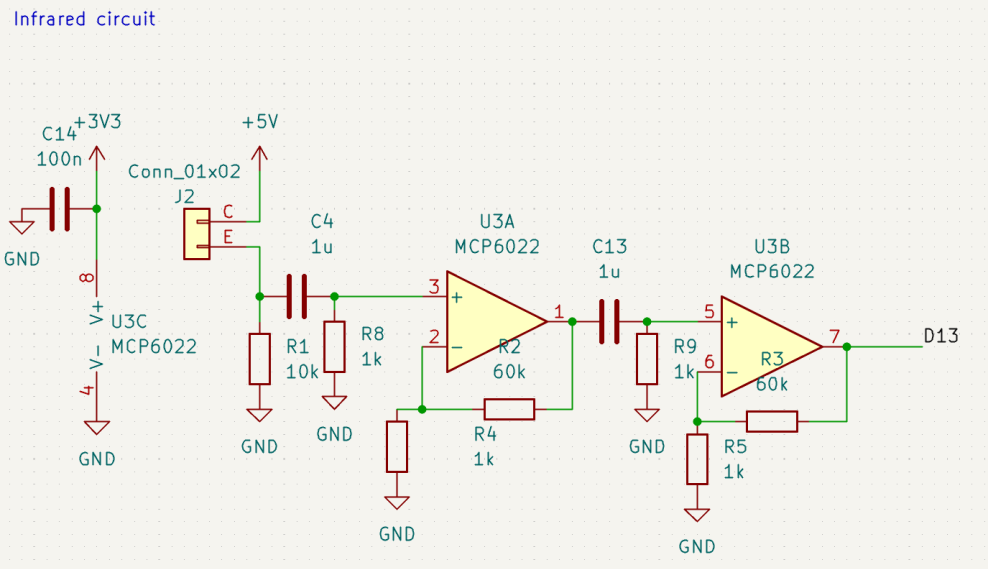


Figure 15: Final infrared sensing-circuit iteration using staged AC coupling, amplification and comparator thresholding.

The detector was also changed after testing. The SFH 300 was usable electrically, but its narrower field of view made detection too dependent on rover alignment and contact angle. The SFH 314 FA was selected because it has a wider acceptance angle, stronger photocurrent options and a spectral peak close to the rock emitter's near-infrared wavelength. This changed the practical requirement from contact-aligned detection to repeatable sensing at approximately 3 cm standoff.

Parameter	**SFH 300**	**SFH 314 FA**
Half-angle sensitivity	50°	80°
Spectral response	Visible and IR	IR-focused
Photocurrent at $V_{CE} = 5\,\mathrm{V}$, $E_e = 1\,\mathrm{W/cm^2}$	at least 3.2 mA	5.6–9.0 mA

Table 8: Phototransistor comparison used to justify replacing the original detector.

2.3.4 Testing and limitations

Bench testing confirmed that the revised front end produced countable pulse clusters at the comparator output. The objective was not to reconstruct the exact optical waveform; it was to generate one clean edge per optical event so that the firmware could apply the Poisson counting rule. A short software refractory interval can reject duplicate interrupts if analogue ringing crosses the threshold more than once.

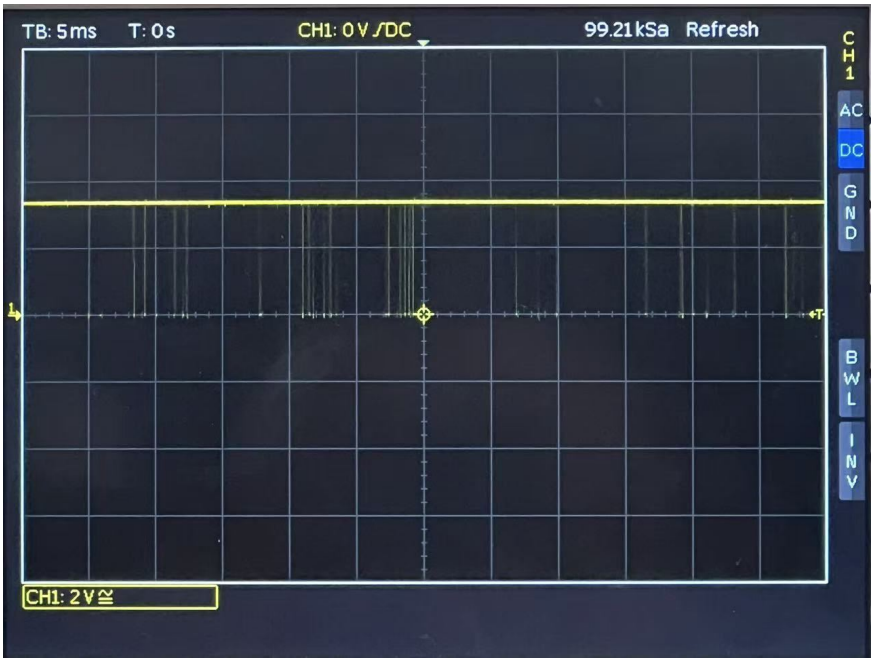


Figure 16: Oscilloscope capture of amplified infrared pulse clusters during prototype testing. Distinct edges support interrupt-based rate counting.

The final validation should record at least 30 count windows for each infrared class with the rock in its realistic covered configuration, then compare the measured means and standard deviations with N_{th}. The remaining weakness is threshold robustness: a fixed comparator threshold is fast and simple, but adaptive baseline rejection would improve immunity to changing laboratory lighting and rover approach angle.

2.4 Magnetic Polarity Detection

The magnetic subsystem determines the binary magnetic orientation of each artificial rock, reported by the rover as *Up* or *Down*. This is a top-level classification requirement because the project brief defines magnetic polarity as one of the three rock-type discriminators, alongside infrared pulse rate and ultrasound response. The subsystem was therefore designed as an interacting hardware–software component rather than a standalone demonstration sensor: it must detect polarity at rover operating distance, reject ambient and self-generated interference, and provide a reliable digital label to the central classification logic.

2.4.1 Requirements Derived from the Brief

Table 9 translates the project-level magnetic requirement into measurable subsystem requirements.

Table 9: Magnetic subsystem requirements and current evidence status.

ID	Requirement	Verification metric	Current status
M1	Determine whether the rock magnet is *Up* or *Down*.	Correct polarity label on a known calibration magnet and on all demonstration rocks.	Bench polarity detection shown.
M2	Operate without requiring physical contact with the rock casing.	Valid polarity reading at practical rover approach distance.	Valid readings observed at 5 cm vertical separation and 1.5 cm horizontal offset; final rover-mounted envelope Pending rover-mounted validation.
M3	Reject ambient magnetic field and invalid sensor values.	Baseline-subtracted reading exceeds a fixed noise threshold; out-of-range values are discarded.	Ambient B_z measured at approximately 22 µT; overflow-like values observed and must be rejected.
M4	Integrate electrically with the rover controller.	Stable I^2C communication at 3.3 V; no bus contention with other peripherals.	Bench integration with Adafruit Metro M0 Express complete..

2.4.2 Field-Strength Estimate and Sensing Envelope

A first-order simulation was used to estimate whether a small embedded magnet could be detected at rover-relevant distances. The model assumed an N35–N45 cylindrical magnet of diameter 8 mm and thickness 3 mm. The simulated field falls rapidly with distance, as shown in Table 10. This justified a sensor-placement requirement: the magnetometer should be placed as close to vertically above the rock as the rover mechanics allow, because the useful signal margin decreases sharply beyond approximately 75 mm.

Table 10: Simulated magnet field strength versus sensor distance. Values in gauss are retained for comparison with magnetic-sensor data sheets; 1 G = 100 μT.

Distance	Field /G	Field /μT	Design implication
30 mm	9.11	911	Strong signal; comfortably above ambient.
50 mm	2.11	211	Suitable target working distance.
75 mm	0.646	64.6	Comparable to the Earth's total field; low margin unless baseline-subtracted.
100 mm	0.277	27.7	Poor margin; not a robust design target.

The Earth's magnetic field is on the same order as the weakest simulated target signals. The

design therefore does not rely on absolute field magnitude. Instead, it measures a local baseline before rock approach and classifies the sign of the baseline-subtracted vertical component, $\Delta B_z = B_z - B_{z,0}$. This is more robust than comparing raw readings against a fixed global threshold, because the rover orientation and local ferromagnetic environment can change the measured background.

2.4.3 Sensor Selection

The selected sensor was the DFRobot SEN0419 breakout, based on the Bosch BMM150 three-axis magnetometer. Its relevant specifications are a 0.3 µT resolution, I^2C/SPI digital interface, 3.3 V operation, and magnetic range of ±1300 µT on the x, y axes and ±2047 µT on the z axis. Table 11 shows why this was selected over simpler magnetic switches.

Table 11: Magnetic sensor trade-off.

Candidate	Sensitivity or resolution	Polarity information	Approx. cost	Decision
Hall-effect switch	Approximately 35 G threshold	Binary, but only after a strong field threshold is crossed	£2.55 for 5	Rejected: threshold is too high for the simulated 3 cm–5 cm operating range.
Magnetoresistive sensor	Approximately 25 G threshold	Insufficient for reliable Up/Down classification in the proposed configuration	£10.00	Rejected: high cost and does not provide the required polarity evidence.
SEN0419/BMM150 three-axis magnetometer	0.3 µT resolution	Signed x, y, z vector components	£5.06	Selected: high signal margin at the target range and direct measurement of vertical polarity.

This selection also improves report-level evaluation quality: it produces continuous quantitative data rather than only a binary switch output, allowing the design to be calibrated, plotted, thresholded, and debugged against measurable noise sources.

2.4.4 Electrical and Software Implementation

The SEN0419 was integrated with the Adafruit Metro M0 Express using I^2C. The four active connections used in the bench prototype are shown in Table 12. The breakout is a 3.3 V device, so the sensor is powered from the controller's 3.3 V rail and not from 5 V. The DFRobot `DFRobot_BMM150` library was used to read the geomagnetic vector via `getGeomagneticData()`, returning signed x, y, and z field components in μT.

Table 12: Bench wiring between SEN0419 and Adafruit Metro M0 Express.

SEN0419 pin	Adafruit Metro M0 Express connection
VCC	3.3 V
GND	GND
SDI	SDA
SCK	SCL

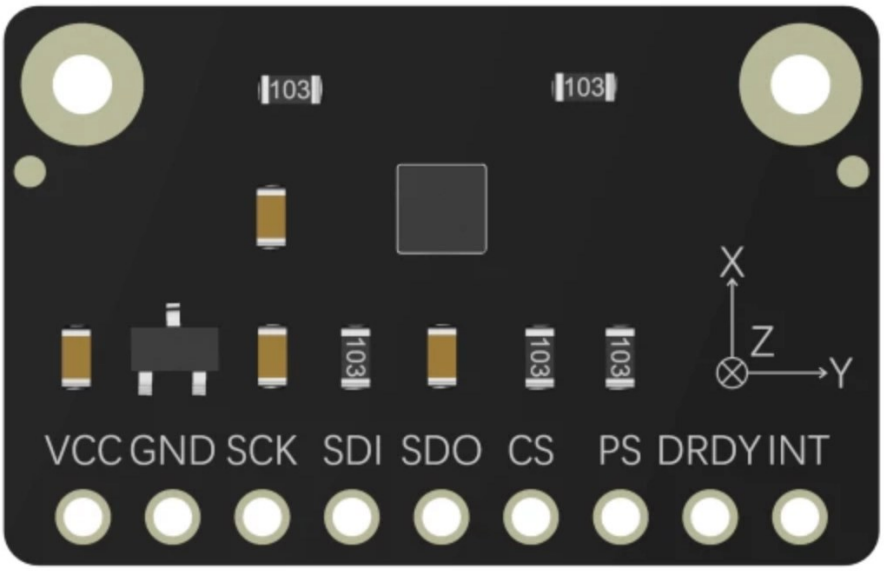


Figure 17: SEN0419/BMM150 breakout showing axis convention and I/O pins.

I^2C was suitable because the sensor data are low-bandwidth compared with the controller clock and the bus provides deterministic master-controlled transactions. For classification, the rover only requires a robust polarity decision rather than high-rate waveform capture.

2.4.5 Bench Testing and Current Evidence

The current bench tests verify that the sensor can measure a meaningful change in vertical field and that polarity can be distinguished under controlled conditions. A magnet was placed approximately 4 cm below the magnetometer, as shown in Figure 18. The observed vertical reading was approximately 130 µT with the magnet present, while the ambient vertical reading was approximately 22 µT. Baseline subtraction therefore gives a target contribution of roughly 108 µT, which is consistent with the simulated order of magnitude at this distance.

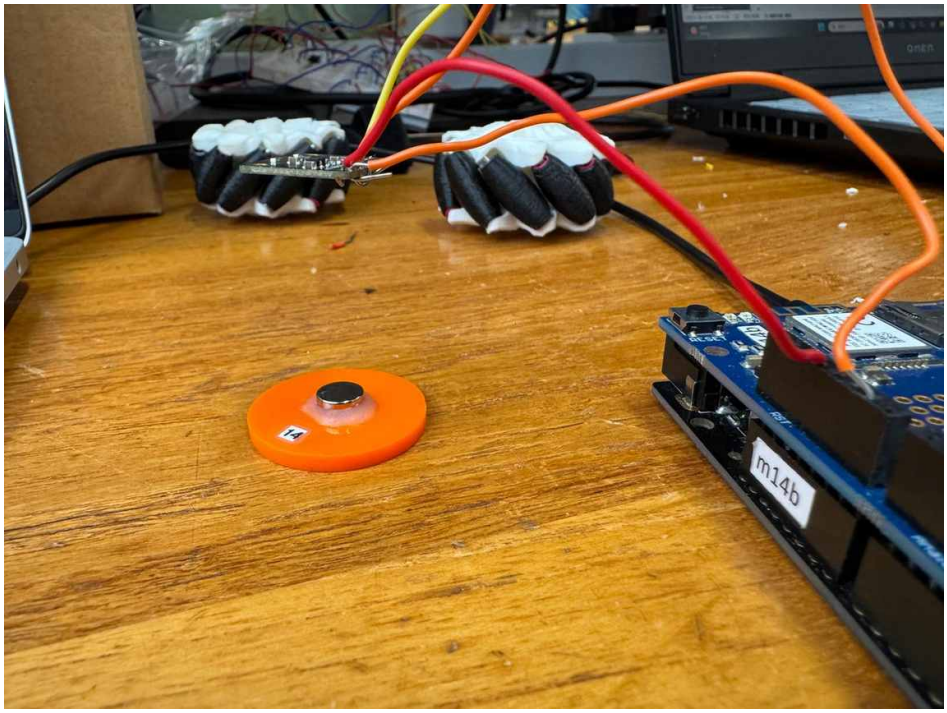


Figure 18: Bench test setup with magnet positioned below the SEN0419 magnetometer.

```
X: 4294967272 uT | Y: 4294967294 uT | Z: 130 uT  → Field UP
X: 4294967272 uT | Y: 4294967294 uT | Z: 128 uT  → Field UP
X: 4294967271 uT | Y: 4294967294 uT | Z: 130 uT  → Field UP
X: 4294967272 uT | Y: 4294967294 uT | Z: 130 uT  → Field UP
X: 4294967272 uT | Y: 4294967294 uT | Z: 129 uT  → Field UP
X: 4294967272 uT | Y: 4294967294 uT | Z: 129 uT  → Field UP
X: 4294967272 uT | Y: 4294967294 uT | Z: 129 uT  → Field UP
X: 4294967271 uT | Y: 4294967294 uT | Z: 129 uT  → Field UP
X: 4294967271 uT | Y: 4294967295 uT | Z: 129 uT  → Field UP
```

(a) Representative field-present readings: $B_z \approx 128\,\mu\text{T}$–$130\,\mu\text{T}$.

```
X: 4294967293 uT | Y: 2 uT | Z: 22 uT  → Weak/ambient
X: 4294967293 uT | Y: 2 uT | Z: 22 uT  → Weak/ambient
X: 4294967293 uT | Y: 2 uT | Z: 22 uT  → Weak/ambient
X: 4294967293 uT | Y: 1 uT | Z: 22 uT  → Weak/ambient
X: 4294967294 uT | Y: 2 uT | Z: 23 uT  → Weak/ambient
X: 4294967294 uT | Y: 2 uT | Z: 23 uT  → Weak/ambient
X: 4294967294 uT | Y: 2 uT | Z: 23 uT  → Weak/ambient
X: 4294967294 uT | Y: 2 uT | Z: 23 uT  → Weak/ambient
X: 4294967294 uT | Y: 2 uT | Z: 23 uT  → Weak/ambient
X: 4294967293 uT | Y: 2 uT | Z: 23 uT  → Weak/ambient
```

(b) Representative ambient readings: $B_z \approx 22\,\mu\text{T}$–$23\,\mu\text{T}$.

Figure 19: Quantitative evidence for baseline subtraction.

A separate polarity test also showed that the sign convention can change consistently when the magnet orientation is reversed. However, Figure 20 shows overflow-like values in one test condition. These values are far outside the BMM150 physical range and must be treated as invalid software or communication artefacts, not as magnetic measurements. This is an important implementation finding: without range checking, a single overflow sample could dominate a threshold rule and cause a false classification.

```
X: 6 uT | Y: 17 uT | Z: 4294967236 uT  → Field DOWN
X: 6 uT | Y: 17 uT | Z: 4294967235 uT  → Field DOWN
X: 6 uT | Y: 17 uT | Z: 4294967235 uT  → Field DOWN
X: 6 uT | Y: 17 uT | Z: 4294967236 uT  → Field DOWN
X: 6 uT | Y: 16 uT | Z: 4294967235 uT  → Field DOWN
X: 6 uT | Y: 16 uT | Z: 4294967236 uT  → Field DOWN
X: 6 uT | Y: 17 uT | Z: 4294967235 uT  → Field DOWN
X: 6 uT | Y: 17 uT | Z: 4294967235 uT  → Field DOWN
X: 7 uT | Y: 17 uT | Z: 4294967235 uT  → Field DOWN
X: 6 uT | Y: 17 uT | Z: 4294967236 uT  → Field DOWN
```

Figure 20: Overflow-like values observed during a polarity test. Final firmware must reject values outside the sensor range before applying the polarity decision.

Table 13 defines the final evidence needed for a high-quality evaluation. The emphasis is on testing both subsystem behaviour and top-level rover behaviour, matching the report marking criterion for evaluation of individual subsystems and the complete system.

Table 13: Magnetic subsystem test matrix

Test	Method	Metric	Outcome
Ambient baseline	Record B_x, B_y, B_z for at least 10 s with no target rock nearby.	Mean, standard deviation, and peak-to-peak noise for each axis.	Initial $B_z \approx 22\,\mu\text{T}$.
Distance response	Measure ΔB_z at fixed vertical separations from 20 mm to 100 mm.	Detection margin $\lvert\Delta B_z\rvert/\sigma_{B_z}$; maximum reliable distance.	5 cm vertical detection observed.
Lateral offset	Sweep sensor laterally across the magnet at the selected probe height.	Maximum offset giving stable Up/Down classification.	1.5 cm horizontal detection observed.
Polarity confusion matrix	Test known Up and Down magnets over repeated trials.	Accuracy, false-Up rate, false-Down rate, no-read rate.	No false polarity decisions observed in preliminary bench trials.
Rover-mounted test	Repeat distance and polarity tests with the sensor mounted on the final chassis or probe.	Change in detection envelope relative to bench test.	Pending rover-mounted validation.
Motor interference	Repeat ambient and polarity tests with motors idle, turning, and stalled/near-stalled.	Change in σ_{B_z}, threshold margin, and classification accuracy.	Pending rover-mounted validation.
Radio-receiver interference	Compare readings with the 89 kHz resonant receiver disabled and enabled.	Difference in mean B_z, noise, and classification accuracy.	Polling at 10 Hz with low-pass filtering; no interference observed in preliminary tests.

Test	Method	Metric	Outcome
Top-level rock classification	Classify all rocks using the full infrared, ultrasonic, radio-age, and magnetic pipeline.	Number of rocks correctly classified; time per rock.	Pending rover-mounted validation.

3 Mechanical and Hardware Integration

The mechanical and hardware design was treated as one integrated platform. The rover had to reach rocks in a constrained arena, hold several directional sensors at repeatable standoff distances, drive below the 750 g weight-sensitive-zone limit, and avoid motor noise corrupting analogue sensing. Mechanical decisions were therefore evaluated against sensing geometry, mass, power distribution and integration risk rather than aesthetics alone.

3.1 Derived Mechanical and Hardware Requirements

ID	Requirement	Verification method	Status
MH1	Rover mass must remain below 750 g.	Final weighing with batteries installed.	In progress
MH2	Rover must manoeuvre between rocks and around obstacles.	Representative arena driving trial.	Validated
MH3	Rover must make small pose corrections near a rock.	Low-speed forward, lateral and rotational tests.	Validated
MH4	Sensor mounts must keep the radio coil, ultrasound transducer, IR detector and magnetometer stable during motion.	Inspection and repeated approach tests.	In progress
MH5	Motor switching must not reset logic or invalidate sensor readings.	Rail and sensor-output checks during driving.	Partially validated
MH6	Four wheels require independent bidirectional control.	Direction and PWM tests on each channel.	Validated
MH7	Layout must fit PCB/stripboard, batteries, Adafruit board, WiFi shield, drivers and sensors.	CAD and physical integration check.	Validated

Table 14: Derived mechanical and hardware requirements used to guide integration.

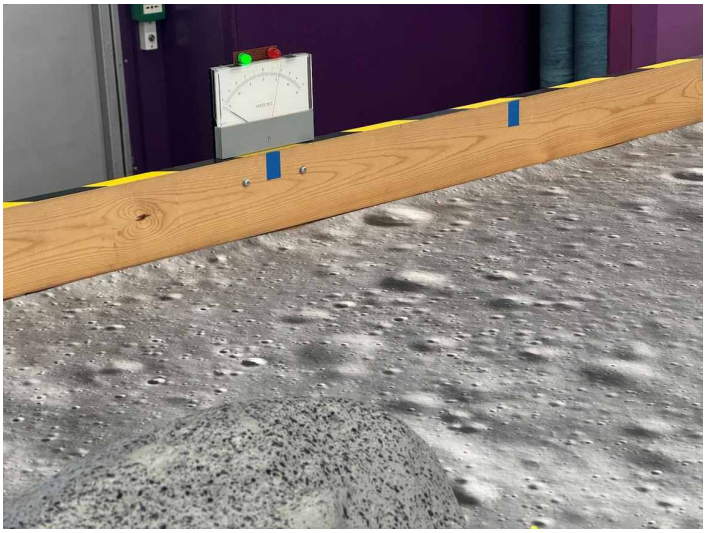


Figure 21: Arena constraints, including rock obstacles and the 750 g weight-sensitive zone.

3.2 Wheel and Drive-System Selection

3.2.1 Drive-system alternatives

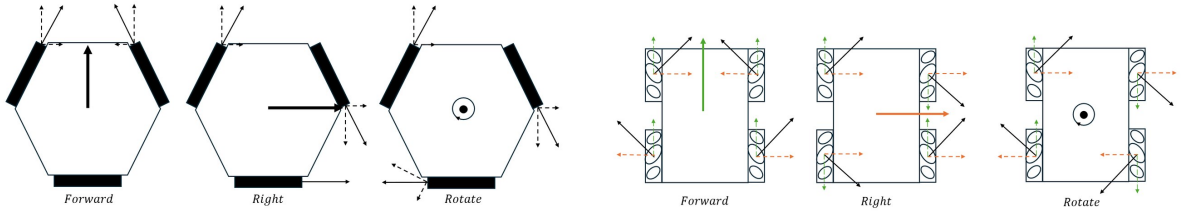
The dominant mechanical requirement was accurate final alignment, not maximum speed. A conventional steering system would require repeated rotations and forward–backward corrections near each rock, increasing the chance that directional sensors would lose alignment. A four-wheel mecanum drive was selected because it allows forward motion, lateral translation and yaw correction without changing the sensor-facing direction.

Drive option	Advantages	Decision
Two-wheel differential	Simple and light.	Rejected: no lateral translation and poor final alignment.
Ackermann steering	Stable forward motion.	Rejected: finite turning radius.
Skid/tank drive	Robust and can rotate approximately in place.	Rejected: relies on slip on the rubber arena surface.
Three-wheel omni	Holonomic with fewer motors.	Rejected: less favourable packaging for rectangular electronics and front sensors.
Four-wheel mecanum	Holonomic, rectangular packaging and direct lateral correction.	Selected despite extra mass and wiring.

Table 15: Comparison of candidate drive systems.

3.2.2 First-order force comparison

A simplified model was used to check that the mecanum choice was technically plausible. If each motor produces torque τ, each wheel radius is r, and the ideal ground force is $T = \tau/r$, then the ideal forward and lateral force for four mecanum wheels is $2\sqrt{2}T$, compared with $\sqrt{3}T$ forward and $2T$ lateral for the three-wheel omni layout. The ideal yaw moment is also higher for the four-wheel layout, $4\sqrt{2}sT$ rather than $3sT$, where s is the effective force-line distance from the centre. The model neglects slip, roller losses and load transfer, but it supports the qualitative decision: the extra wheel and driver complexity bought better controllability in the directions most relevant to close-range sensing.



(a) Three-wheel omni force model. **(b)** Four-wheel mecanum force model.

Figure 22: First-order drive-force comparison used during drivetrain selection.

3.2.3 Wheel diameter selection

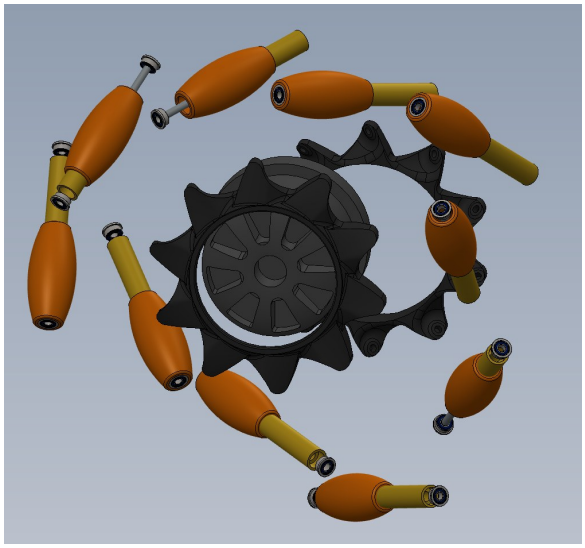
The 60 mm wheel diameter balanced clearance, speed and torque demand. Since $F = \tau/r$, larger wheels reduce available ground force for a fixed motor torque, while smaller wheels reduce ground clearance and make the rover more sensitive to surface irregularities and cable drag. The selected size gave adequate clearance while still permitting low-speed movement increments below the target 10 mm.

Criterion	Target	Measured	Result
Minimum forward displacement in 0.1 s	10 mm	5.5 mm at PWM 100	Pass
Minimum lateral displacement in 0.1 s	10 mm	5.0 mm at PWM 100	Pass
Speed over 500 mm at PWM 255	200 mm s^{-1}	250 mm s^{-1}	Pass
Straight-driving current	< 0.6 A	0.123 A	Pass
Lateral/turning current	< 0.6 A	0.121 A	Pass

Table 16: Validation evidence for the 60 mm mecanum-wheel selection. All readings are averages of ten measurements.

3.2.4 Mecanum wheel manufacture and iteration

The mecanum wheels were manufactured in-house to avoid additional cost and to match the available chassis geometry. The design used ten angled TPU rollers per wheel, PETG supports and caps, and an initially printed shaft/bearing concept. A trial print showed that the TPU roller interference fit worked, but the 1.8 mm printed shafts and fine bearing features were below the reliable resolution of the laboratory printer: the 0.4 mm nozzle produced an effective bead of about 0.6 mm, causing rough, weak shafts.



(a) Exploded CAD view.



(b) Completed wheel.

Figure 23: Wheel iteration from CAD design to final assembly using metal mandrel shafts.

Failure mode	Evidence	Design change	Outcome
Printed shaft fracture	1.8 mm shaft was too fine for the printer resolution.	Replace with 1.9 mm rivet mandrel.	Passed assembly.
Poor bearing definition	Printed bearing geometry did not rotate reliably.	Remove bearing; use shaft/roller clearance.	Simpler build.
Short first shaft cut	First mandrel did not locate in both caps.	Cut rivet pin first, then trim and file.	Repeatable length.
Grip uncertainty	Roller needed compliance without collapse.	TPU roller over PETG support.	Good fit and traction.

Table 17: Wheel-manufacturing failure modes and design responses.

3.3 Power Distribution and Motor Control

3.3.1 Motor-drive requirement

Four 6 V DC motors cannot be driven directly from the Metro M0 pins because GPIOs provide logic signals, not motor current or polarity reversal. Two TC78H620FNG dual H-bridge drivers were therefore used, giving four independent bidirectional channels. Their protection features—under-voltage lockout, over-current protection, short-circuit protection and thermal shutdown—also reduced the consequence of integration faults during bench testing.

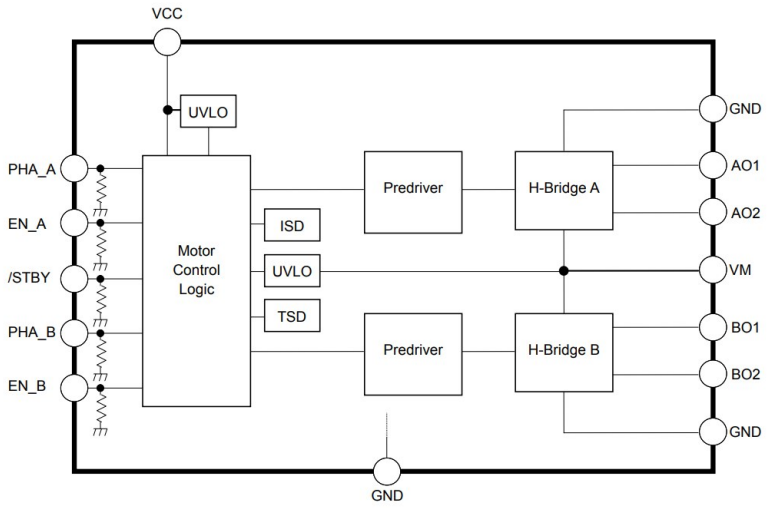
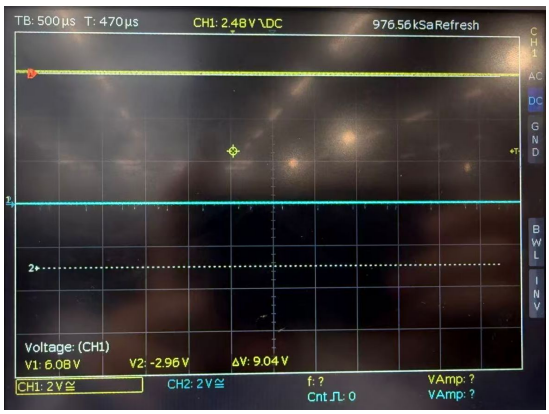


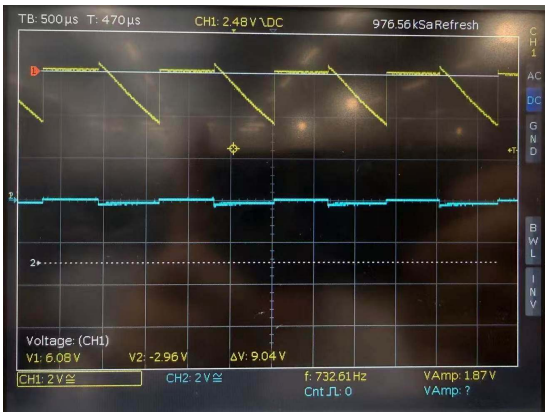
Figure 24: TC78H620FNG dual H-bridge block diagram used for the fallback motor-control design.

3.3.2 Direction and PWM speed control

Direction pins select motor polarity, while enable pins receive PWM. For duty cycle D, the approximate average motor voltage is $V_{\text{motor,avg}} \approx DV_M$. Oscilloscope checks confirmed that duty-cycle changes propagated from the microcontroller to the driver outputs, validating the electrical control path before full driving tests.



(a) PWM = 100 %.



(b) PWM = 50 %.

Figure 25: Representative oscilloscope waveforms confirming PWM switching behaviour.

3.3.3 Power regulation and hardware safety

The motor rail was kept separate from the regulated logic rail to avoid motor transients resetting the microcontroller or shifting sensor thresholds. The driver motor input is powered from the nominal 6 V rail, while logic is compatible with 3.3 V control signals. Measured operating currents were well below the 0.8 A per-channel operating limit, leaving margin for startup and short mechanical disturbances.

Condition	Measured current per motor	Driver limit	Margin
No-load motor spinning	0.135 A	0.8 A operating	0.665 A
Straight driving	0.123 A	0.8 A operating	0.677 A
Lateral mecanum motion	0.121 A	0.8 A operating	0.679 A
In-place rotation	0.121 A	0.8 A operating	0.679 A
Blocked-wheel check	0.180 A	1.0 A absolute max	0.820 A

Table 18: Motor-driver current-margin validation.

Test	Target metric	Measured result	Pass?
Logic rail during motor startup	No reset within supply tolerance.	No reset observed.	Y
Motor rail during reversal	Remain within driver range.	7 V	Y
PWM duty-cycle check	Within 10 % of command.	6 % error	Y
Driver temperature after continuous driving	Safe after 5 min.	Safe to touch.	Y
Sensor output with motors on/off	No classification-relevant shift.	To be tested.	TBD

Table 19: Power and noise validation tests for the motor-control subsystem.

3.3.4 Motor-driver connections and pin allocation

Pin allocation was revised around WiFi-shield reservations and sensor inputs. In particular, pins 5, 7 and 10 could not be used freely. The rear-right direction signal was moved from D5 to D2, and the front-right enable signal was moved from D7 to D12.

Driver	Motor	Direction pin	PWM pin	Note
D1	M_{FL}	D8	D11	Front-left motor.
D1	M_{FR}	D13	D12	Enable reassigned from D7.
D2	M_{RL}	D4	D6	Rear-left motor.
D2	M_{RR}	D2	D3	Direction reassigned from D5.

Table 20: Final motor-driver pin allocation after WiFi-shield and sensor conflicts were resolved.

3.3.5 Supply-chain delay and fallback decision

The original WPSH207 dual-L293D motor-driver shield was attractive because it integrated four channels on an Arduino-style board. However, the delivery estimate moved from 5 June to 28 June, after the useful integration window. The team therefore changed to the available TC78H620FNG drivers. This increased wiring complexity but reduced the larger risk of reaching system testing without a functioning drivetrain.

Criterion	WPSH207 shield	TC78H620FNG fallback
Availability	Delayed beyond useful integration window.	Available immediately.
Integration complexity	Lower: single shield and terminal blocks.	Higher: two drivers and custom wiring.
Schedule risk	High after estimate moved to 28 June.	Lower; testing could continue.
Pin flexibility	Less adaptable.	Reassigned around WiFi and sensor conflicts.
Decision	Rejected due to delivery risk.	Selected to protect integration time.

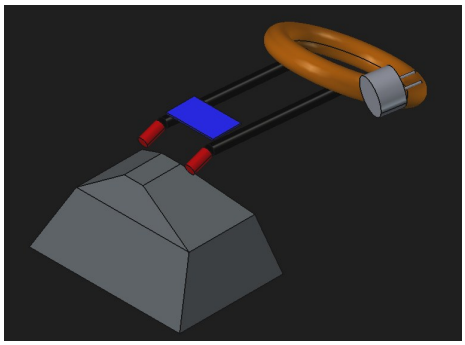
Table 21: Fallback decision after the original motor shield became unavailable.

3.4 Chassis Iteration and Mechanical Design

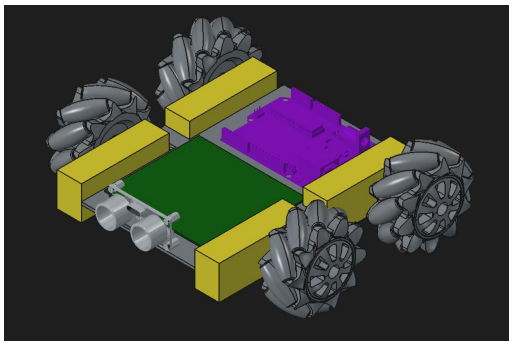
The chassis had to locate the drivetrain rigidly, support the batteries and control electronics, and preserve sensor geometry. FreeCAD was used so packaging and mass distribution could be revised as electrical constraints changed.

3.4.1 Iteration 1: single-layer chassis

The first design was a single-tier 120 mm by 150 mm chassis. It was rejected because the full width with motors and mecanum wheels exceeded about 230 mm, reducing clearance between obstacles, and because the front sensors would have been supported by a glued extension with poor stiffness. That was unacceptable for the radio coil and ultrasonic transducer, whose outputs depend on angle and standoff distance.



(a) Initial front sensor concept.

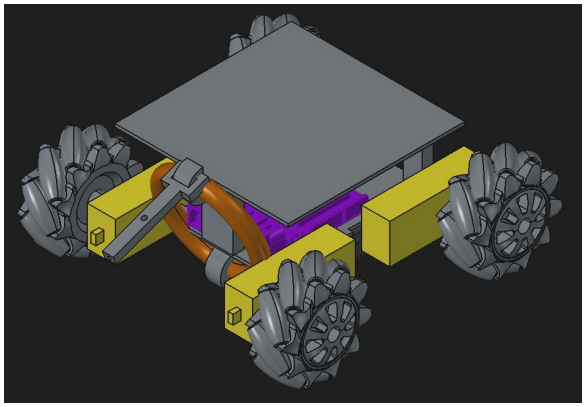


(b) Rejected single-tier chassis.

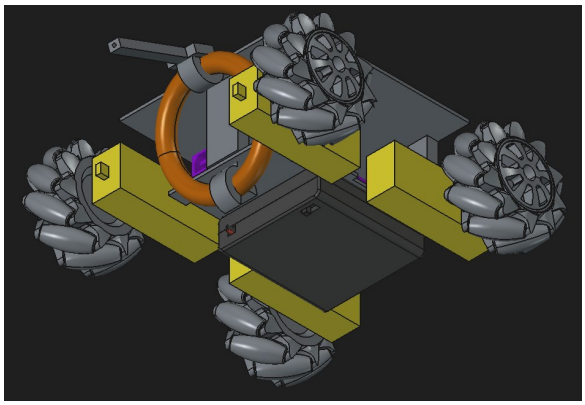
Figure 26: First chassis iteration and rejected sensor-placement approach.

3.4.2 Final dual-layer chassis

The final architecture uses a lower layer for heavy/noisy components—motors, batteries, supports and the control board—and an upper layer for sensing and circuit modules. This reduces footprint, lowers the centre of mass, and separates analogue sensing from motor wiring. Front sensor clamps with 10 mm internal diameter locate the radio coil and reduce alignment drift during repeated approaches.



(a) Front/isometric view.



(b) Rear/isometric view.

Figure 27: Final dual-layer chassis design.

3.4.3 Models, dimensions, and mass allocation

The bottom layer is 70 mm by 100 mm and carries the drivetrain and battery mass; the top layer is 112 mm by 110 mm and provides space for the sensor and circuit boards. The measured subtotal of major components was 489.9 g, leaving 260.1 g for chassis parts, wiring, fasteners, motor drivers and mounting hardware before the 750 g limit.

Component	**Mass / g**
Sensors	14.6
Motors	119.8
Wheels	197.2
Adafruit board	20.7
WiFi shield	18.6
Batteries	$26.1 \times 4 = 104.4$
Battery holder	14.6
Measured subtotal	**489.9**
Remaining allowance before 750 g limit	**260.1**

Table 22: Mass budget for the main known components.

3.5 Hardware Prototyping Evolution: Breadboard, Stripboard, and PCB

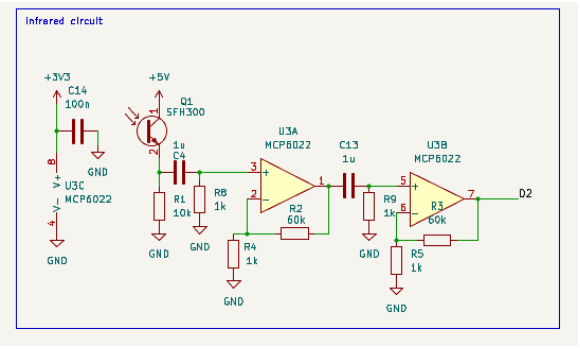
3.5.1 Breadboard prototype

Breadboards were used only for early circuit proof-of-concept testing, where component values and signal-conditioning assumptions could be changed quickly. They were rejected for final integration because loose jumper contacts, vibration and footprint would make them unreliable on a moving mecanum platform.

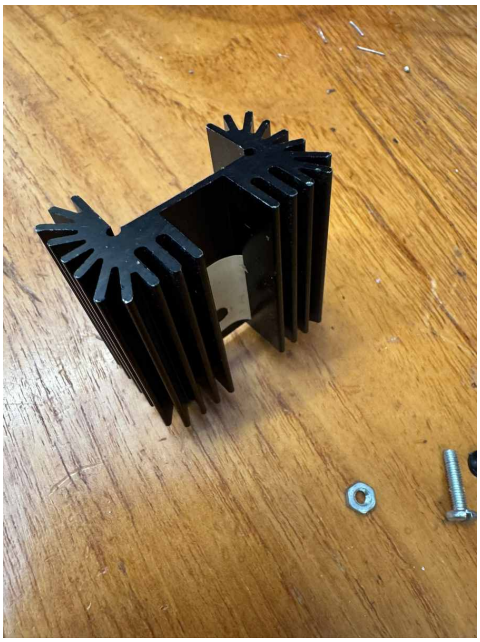
3.5.2 Stripboard implementation and thermal revision

Stripboard was evaluated as an intermediate route. It was robust enough for soldered prototypes and allowed staged debugging, but routing became congested once analogue sensing, power rails and connectors were combined. The regulator also overheated during sustained operation, causing the 5 V rail to fall after about one minute. Later fault-finding showed that the root cause was not the regulator package itself but excessive load current caused by an unintentionally

low-resistance path in the circuit. Replacing this resistance with the intended higher value reduced the current draw, stabilised the 5 V rail and removed the thermal fault. The separate 40 g aluminium heat sink was therefore useful as a diagnostic workaround, but it was not carried into the custom PCB design.



(a) IR stripboard planning.



(b) Regulator heat-sink workaround.

Figure 28: Stripboard planning and thermal revision.

3.5.3 Custom PCB solution

The final board was a 6 cm by 10 cm two-layer through-hole PCB designed in KiCad. Through-hole components were chosen for manual assembly, value changes during calibration and reduced risk through IC sockets. Connectors were placed at board edges, op-amp stages were grouped by signal flow, decoupling capacitors were placed near IC supply pins, and a B.Cu ground pour reduced ground-track congestion. Tracks were routed at approximately 0.25 mm, with local power paths increased to about 0.4 mm. KiCad DRC reported no unconnected nets or clearance violations before Gerber export.

The final schematic consolidated the radio, infrared and ultrasonic analogue chains with the Arduino input/output connectors and the regulated power interface. Capturing all interfaces in one KiCad design reduced the risk of late wiring errors because pin names, supply rails and signal directions could be checked before layout.

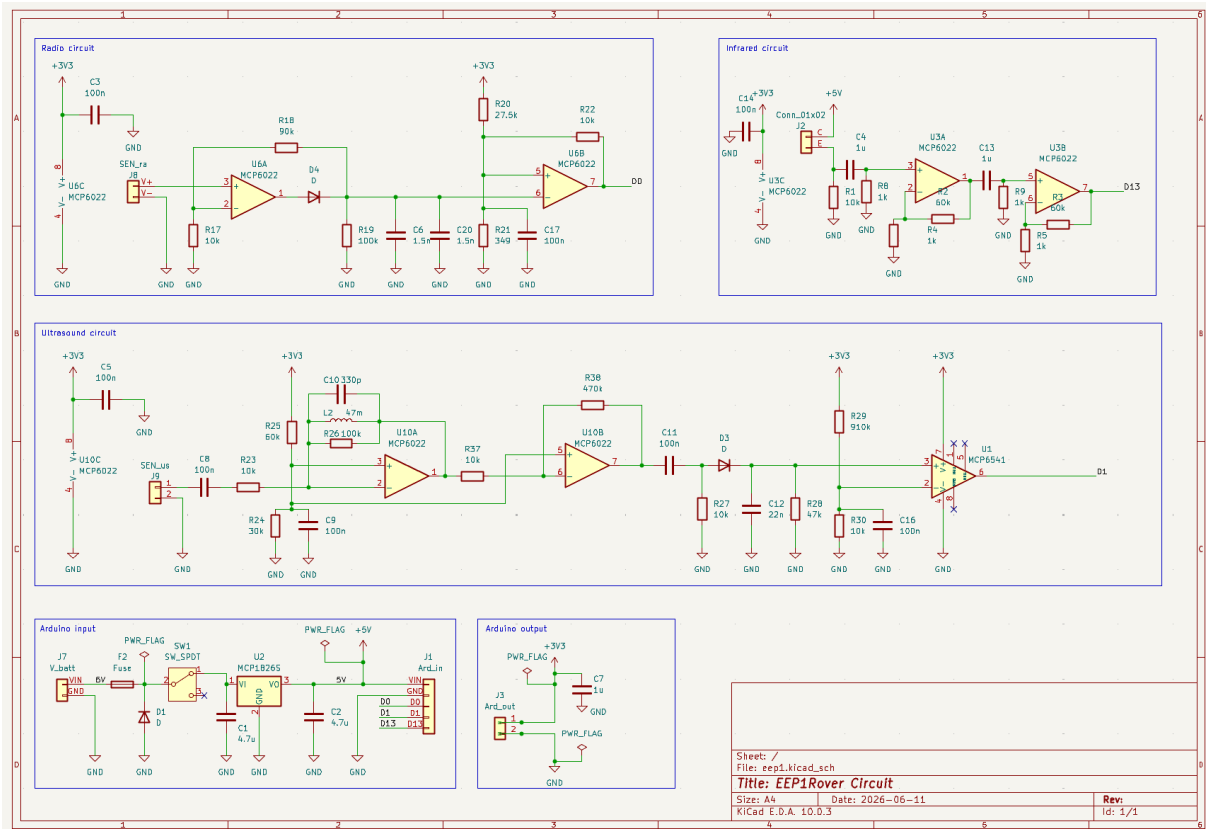
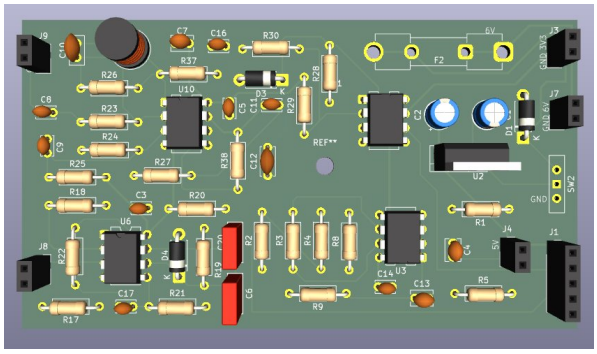
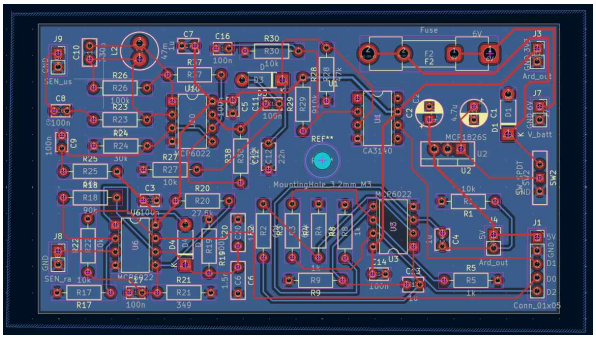


Figure 29: Final consolidated KiCad schematic for the rover interface PCB, including radio, infrared, ultrasonic, power and Arduino connector sections.



(a) PCB 3D front view.

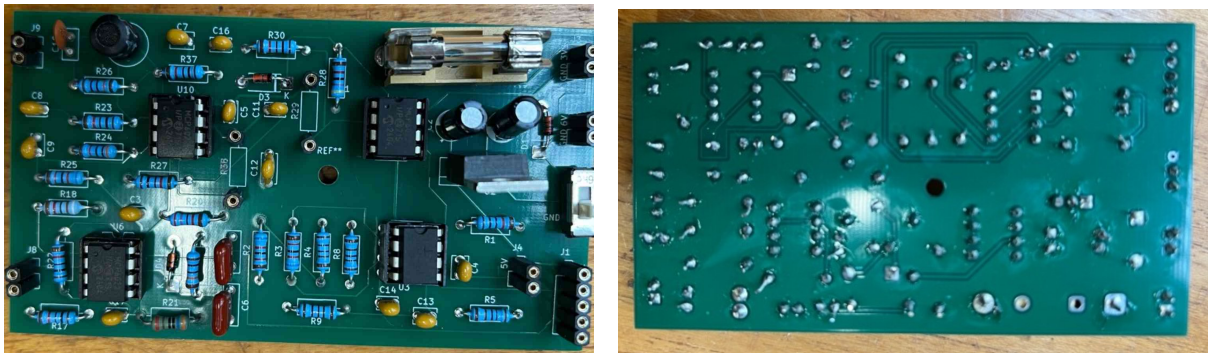


(b) Final routing view.

Figure 30: Custom PCB implementation for compact and repeatable sensor integration.

3.5.4 PCB assembly and testing

The completed PCB was inspected on both sides before powered tests. 8-pin IC sockets were used for all four ICs, protecting the op-amps and comparator from soldering heat and allowing replacement during debugging. One-by-one pin sockets were fitted at selected resistor positions so calibration values could be changed without re-soldering.



(a) Populated top side. **(b)** Solder side.

Figure 31: Final PCB assembly used for solder-joint inspection, socketed-IC validation and resistor-value adjustment.

Initial power-up showed regulator heating. Subsequent fault-finding traced this to excessive current from an unintentionally low-resistance path; replacing it with the intended higher resistance reduced load current and stabilised the regulated rail.

Medium	Strength	Limitation	Decision
Breadboard	Fastest for value tuning.	Heavy, bulky and unreliable under vibration.	Proof-of-concept only.
Stripboard	Soldered and compatible with large heat sink.	Routing congestion and thermal penalty.	Retained only where needed.
Custom PCB	Lowest wiring complexity and best robustness.	Less flexible after manufacture.	Selected for final signal integration.

Table 23: Comparison of physical prototyping media.

4 Software, Control and Integrated Evaluation

The software and control layer converted the independently developed sensing circuits into a single operator-facing rover. Its main design requirement was not only to read each subsystem, but to make those subsystems cooperate under realistic timing, communication and failure conditions. The final architecture used the Adafruit Metro M0 Express as the embedded controller, digital interfaces wherever possible, WiFi telemetry for the dashboard, and a fusion algorithm that could still classify a rock when one type sensor was missing or unreliable.

4.1 Interface contract and firmware architecture

The key integration decision was to define a digital contract between hardware and software early. The radio receiver produced a UART age packet, the infrared circuit produced a pulse train, the ultrasonic circuit produced a binary detection input, and the BMM150 magnetometer used I^2C. This limited the amount of analogue interpretation performed in firmware and made each subsystem testable in isolation before the rover was fully assembled.

Subsystem	Electrical interface	Firmware interpretation	Failure handling
Radio age	`Serial1` UART at 600 bit/s	Parse four-character packets such as `#123`; convert to billions of years.	Reject malformed packets and retain the previous valid age until a new valid packet is received.
Infrared type	Digital pulse input	Count pulses in a fixed window and classify rate statistically.	Report the channel invalid if the count is outside plausible limits or no stable rate is present.
Ultrasound type	GPIO logic input	Detect the presence of a 40 kHz ultrasonic emission.	Treat missing or unstable logic as an unavailable bit rather than forcing a rock type.
Magnetic polarity	BMM150 over I^2C	Read the signed magnetic vector and classify upward or downward polarity.	Use a dead-zone around ambient field and mark the channel invalid when the reading is weak.
Dashboard/control	HTTP over WiFi	Exchange motor commands and JSON telemetry with the browser interface.	Apply a stop state if commands time out or the connection becomes stale.

Table 24: Digital interface contract used to reduce late-stage integration ambiguity.

The firmware was structured around a non-blocking main loop. Sensor updates, classification, motor command handling and telemetry serving were scheduled without long delays so that the rover remained responsive while collecting data. This mattered because blocking code in one subsystem would have made the WiFi dashboard lag, delayed motor stop commands and distorted the fixed counting window used by the infrared classifier. A simulation mode was also retained. It allowed the dashboard, fusion logic and motor command parser to be tested with placeholder sensor values before every analogue front end was available.

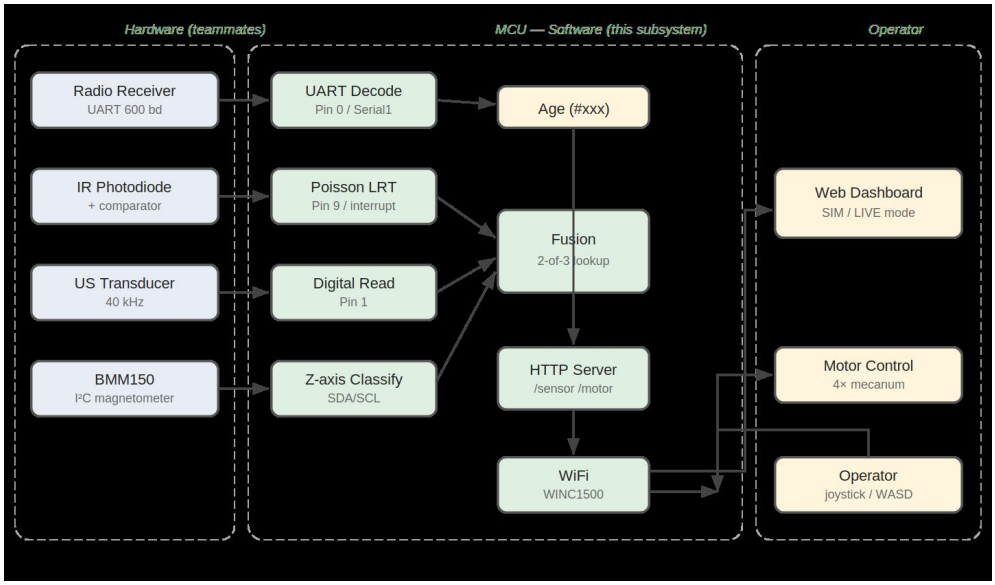


Figure 32: Integrated data flow from physical sensors through firmware classification, fusion and WiFi telemetry to the operator dashboard.

4.2 Age decoding and type fusion

The age channel was treated separately from rock type. The radio circuit demodulated the 89 kHz ASK signal and passed a serial packet to the Metro M0. The parser accepted only correctly framed packets, reducing the chance that comparator noise or an inverted receiver level would be converted into a false age. When a valid packet was received, the three digits were converted into a decimal age, for example `#123` becoming 1.23 billion years.

Rock type was determined by three binary physical properties: high or low infrared rate, ultrasonic presence or absence, and magnetic polarity. The firmware stored these as separate sensor bits before applying the lookup table. The useful property of the project encoding is that any two of the three type bits identify a unique rock. The fusion algorithm therefore did not need to fail immediately when one sensor was unavailable. If at least two type channels were valid and their combination matched a unique rock, the rover reported the corresponding type; if fewer than two were valid, or if the available combination was inconsistent, it reported `UNKNOWN`.

Rock type	Infrared rate	Ultrasound	Magnetic polarity
Basaltoid	High	Present	Down
Gravion	Low	Absent	Down
Regolix	Low	Present	Up
Lunarite	High	Absent	Up

Table 25: Rock-type lookup table implemented in firmware. The encoding makes each two-sensor combination unique.

This design choice improved robustness and reportability. A missing magnetometer reading, weak ultrasonic coupling or temporary IR dropout did not automatically create a wrong answer. Instead, the user interface could distinguish between a confident classification, a degraded but still valid classification, and an unknown result requiring another measurement.

4.3 Infrared classification algorithm

The firmware implements the decision rule derived in Section 2.3. Each comparator edge increments an interrupt-driven counter during a fixed observation window. For the nominal 0.5 s window the decision threshold is 215 counts: values at or above this are classified as the high-rate infrared class, and lower values as the low-rate class. This method is more robust than measuring a single inter-arrival time because the source is stochastic and individual optical events may be missed or duplicated by analogue ringing.

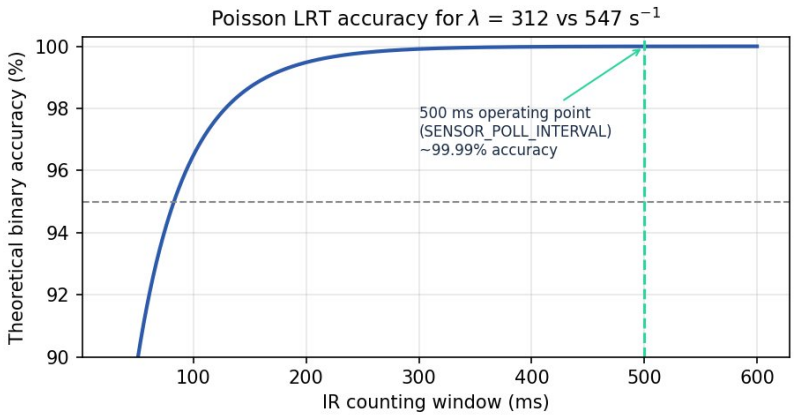


Figure 33: Expected infrared classification accuracy for a 500 ms counting window, showing sufficient separation when the analogue front end produces clean digital pulses.

The analogue and software designs are therefore matched: hardware converts short optical pulses into logic-level events, while firmware performs statistical averaging over the window. If testing shows excessive missed edges under ambient light or poor alignment, the same code can increase the window length without changing the interface contract.

4.4 WiFi dashboard and rover control

The operator interface was implemented as a browser dashboard served over WiFi. It displayed the latest rock age, rock type, sensor-bit state and connection status, while also sending movement commands to the rover. This avoided a separate desktop application and made the interface usable from any device connected to the rover network. The dashboard and firmware communicated through lightweight HTTP requests and JSON telemetry, which were simple enough for the Metro M0 and WiFi module while still being readable during debugging.

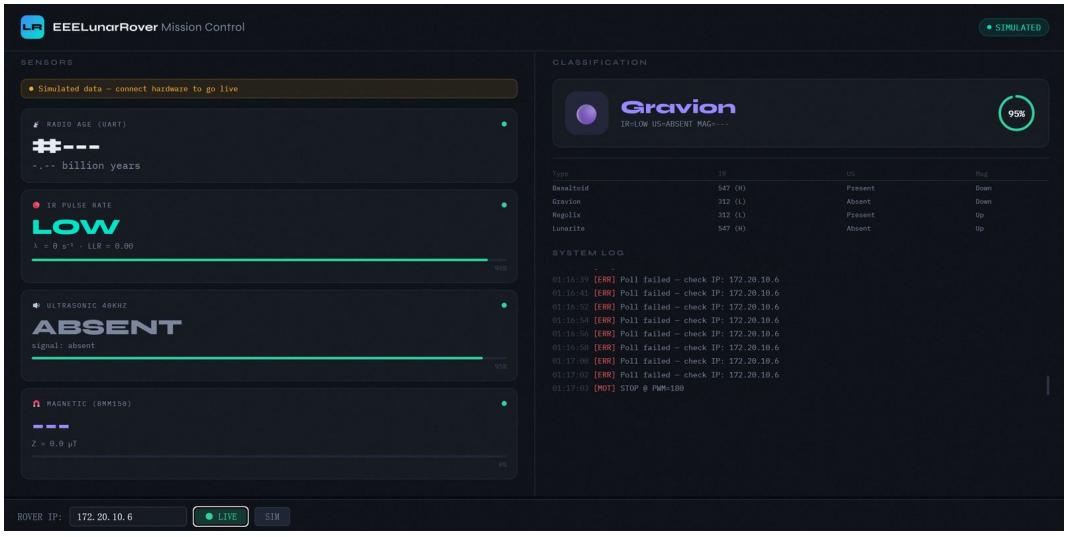


Figure 34: Dashboard evidence from live-style telemetry, showing operator controls and a decoded rock classification.

Motor control was designed for a mecanum-wheel chassis rather than a conventional differential drive. The command layer therefore mapped operator inputs to front-left, front-right, rear-left and rear-right wheel directions so the rover could translate sideways, rotate and move diagonally. Multiple speed modes were supported to balance fast traversal against precise alignment during

sensing. A fail-safe stop behaviour was included so that stale commands did not leave the rover driving after a communication interruption.

4.5 Integrated evaluation

The evaluation strategy followed the marking requirement that both subsystem and top-level behaviour should be tested. The available evidence showed strong subsystem progress and partial full-system integration. The IR pathway had a complete design route from analogue pulse generation to statistical classification. The magnetometer had I^2C implementation and polarity testing, with weak-field limitations identified. The dashboard and WiFi layer had live telemetry evidence. The motor mapping was implemented for mecanum control. The remaining gap was a complete arena test set in which every rock was approached repeatedly with the final hardware stack and the final dashboard recording age, type and motion response.

```
Connecting to WPA SSID: iPhone
HTTP server started @ 172.20.10.5

System ready. Waiting for sensor data...

[IR] Counting window started, waiting for first interval...
[US] 40kHz PRESENT (conf=0.95)
```

Figure 35: Serial/WiFi diagnostic output used to verify telemetry exchange and firmware state reporting during integration.

Requirement	Evidence available in the final records	Status
Decode radio age	UART packet parser and radio receiver tests; malformed packets rejected by framing logic.	Partially verified
Classify IR rate	Analogue pulse front end, count-window classifier and theoretical error analysis for the two specified rates.	Strong subsystem evidence
Detect ultrasound	40 kHz detection circuit and digital interface included in the hardware/software contract.	Subsystem evidence
Detect magnetic polarity	BMM150 implementation, polarity readings and dead-zone handling; limited by sensor distance and magnet strength.	Partially verified
Fuse rock type	Lookup table and two-sensor fallback implemented; returns `UNKNOWN` instead of forcing an unsupported result.	Strong software evidence
Remote control	Browser dashboard, telemetry display and mecanum command mapping implemented.	Integrated evidence
Top-level rover performance	Chassis, power, sensing and dashboard integrated, but final repeated arena data were not fully recorded.	Further test required

Table 26: Integrated evaluation summary and remaining evidence gaps.

The most important limitation is therefore evidential rather than architectural. The report contains measured and bench-tested subsystem results, but the final mark would be better supported by a compact arena test matrix: each rock measured at multiple orientations and distances, with the dashboard recording age, IR bit, ultrasound bit, magnetic bit, final type, motion command responsiveness and any unknown classifications. This would convert the present engineering evidence into a complete requirement-by-requirement acceptance test.

4.6 Conclusion

The final integrated design is a technically complex first-year rover rather than a collection of independent circuits. Its strongest features are the clear digital interface contract, the statistical treatment of IR pulse rates, the two-sensor fusion fallback, the WiFi dashboard and the mecanum control layer. The main unresolved weakness is the absence of a fully repeated final arena data set. Nevertheless, the system demonstrates a logical progression from project requirements to implementation and evaluation: physical signals are conditioned by the hardware, interpreted by firmware, fused into a rock classification and presented to an operator through a remote-control interface.

Professional Reflection Forms

First Name: Fred
Last Name: Ayres
CID: 02657015
Group Number: 14

ELEC40006-Electronics Engineering Design Project 1

Reflection on Professional Practice Form

Please answer all of the following questions in a total of 300 words (+10%). This is a compulsory component of the module and must be passed.
Include your name, surname, and CID number on the form.
Each group member must complete the Reflection on Professional Practice Form individually.
All completed forms must be compiled and attached to the end of your group's project report.
Submit the full report, including all reflection forms, by 17:00pm on 11 June 2026.

1. How effectively did your group function as a team? How effective was your personal contribution to the team?
2. How did a diversity of people in the teams help you to perform better as a team?
3. What skills and knowledge did you acquire in this project? How?

1. I am pleased that everyone contributed to at least one sensing circuit, although some people were able to contribute more to the team. I think we made a good effort to distribute the work fairly given our time constraints. For example, when one of us was struggling with one of the sensor circuits, other people in the team volunteered to help such that we would deliver on time. Personally, I worked on research and testing of the Magnet sensor, Design and Implementation of the IR circuit, Power System Design and Strip Board Construction.

2. A diversity of people in the teams allowed everyone to work on different sections on the project, working at their strengths and limitations.

3. I significantly improved my soldering skills as I was responsible for soldering the entire stripboard. I improved my proficiency in designing detection circuits where I had to consider different sources of noise (e.g. ambient DC noise from room lighting or Johnson Noise due to a very large gain stage in our circuit), find ways to reduce it: filter circuits. I also had to consider the gain in my circuit given the limitations: I wanted to be able to detect my rock at least 3cm away but if my gain was too large it would produce a lot of noise. Therefore, I also had to assist in researching a new IR sensor with a larger angle of detection because our previous sensor had a limited field of view. As we initially planned to use different drivers (the WPSH207 driver shield) for our circuit, I was responsible for researching and designing the new power system. Although, in the end due to our shield not arriving in time, we ended up using the drivers given to us.

First Name: Sijie
Last Name: Fei
CID: 02686898
Group Number: 14

ELEC40006-Electronics Engineering Design Project 1

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2. How did a diversity of people in the teams help you to perform better as a team?
3. What skills and knowledge did you acquire in this project? How?

1. Our group of six divided into hardware and software streams. I owned the entire firmware and web dashboard: UART age decoding, IR classification using a Poisson log-likelihood ratio test, ultrasonic and magnetometer processing, multi-sensor fusion, mecanum motor control, and the real-time operator interface. Teammates designed the analogue front-ends — radio receiver, IR detector, ultrasonic interface, and motor driver wiring. We agreed a digital interface contract early (signal types, pin assignments, voltage levels), enabling both streams to develop in parallel. I built and tested the full classification pipeline in simulated mode before any hardware existed, so integration became a matter of flipping a compile flag. When the radio receiver turned out to have inverted logic levels, having working software made the fault easy to isolate. My contribution was substantial in scope but narrow in discipline; attending more hardware bench sessions would have helped me catch interface issues earlier.

2. Our team spanned different nationalities and academic strengths. Members with stronger analogue-electronics backgrounds designed robust sensor circuits, while my programming experience enabled the classification algorithms and dashboard. One teammate with mechanical experience led the chassis modifications for mecanum wheels. This range meant no single point of failure — someone could always offer a different perspective when another was stuck. Communicating across disciplines also forced precise interface documentation, which improved overall system quality.

3. I learned ARM embedded programming on the SAMD21, non-blocking real-time architecture, and HTTP server design under severe resource constraints — the WiFi101 module supports only four concurrent sockets. Implementing the Poisson LRT taught me to apply statistical decision theory to a real classification problem. The dashboard required asynchronous JavaScript, touch-event handling, and responsive design. Most valuably, debugging a motor-control watchdog failure taught me that embedded networking differs fundamentally from desktop networking: latency and concurrency assumptions that hold on a laptop collapse on a microcontroller. These lessons came through iterative development and integration failures no textbook could replicate.

First Name: Zeyu
Last Name: Liu
CID: 02590274
Group Number: 14

ELEC40006-Electronics Engineering Design Project 1

Reflection on Professional Practice Form

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1. How effectively did your group function as a team? How effective was your personal contribution to the team?
2. How did a diversity of people in the teams help you to perform better as a team?
3. What skills and knowledge did you acquire in this project? How?

1. Overall, our group worked reasonably. We divided tasks based on each member's strengths and interests, communicated about the main design issues, and combined our work into the final project and report. However, the teamwork was not perfect, as some members contributed less consistently than others, creating uneven workload and slowing down progress. My personal contribution was mainly in PCB design/construction/testing, ultrasonic signal detecting circuit, circuit prototyping/debugging, and presentation/report writing. I was able to complete my work efficiently and effectively.

2. The diversity within the team helped because different members had different strengths and approaches. Some focused more on circuit behaviour and component choices, while others considered layout, testing, assembly, or documentation. This helped the active members identify problems from more than one perspective. For example, different opinions on sensor placement, component choice, and PCB manufacturability were considered along the PCB design. However, the benefit of diversity was limited by the unequal level of participation across the group.

3. I gained practical knowledge of PCB design, sensor integration, circuit design, circuit construction, manufacturing preparation, and circuit/ PCB debugging. I developed a better understanding of how an engineering design project moves from an initial idea to a practical working system. I improved my ability to apply technical knowledge to real designs, rather than only solving isolated theoretical problems. I also improved skills in documentation, and group communication. All above were developed through working on the design, solving practical problems during the project, responding to issues found during testing, and contributing to the project consistently and showing up everyday.

First Name: Owais
Last Name: Darwesh
CID: 02568029
Group Number: 14

ELEC40006-Electronics Engineering Design Project 1

Reflection on Professional Practice Form

Please answer all of the following questions in a total of 300 words (+10%). This is a compulsory component of the module and must be passed.
Include your name, surname, and CID number on the form.
Each group member must complete the Reflection on Professional Practice Form individually.
All completed forms must be compiled and attached to the end of your group's project report.
Submit the full report, including all reflection forms, by 17:00pm on 11 June 2026.

1. How effectively did your group function as a team? How effective was your personal contribution to the team?
2. How did a diversity of people in the teams help you to perform better as a team?
3. What skills and knowledge did you acquire in this project? How?

1. I think overall our group functioned well as a team. Everyone was very early split off into the sub-modules we decided early on. It did mean working in isolation for periods of time, however I think it suited many members of our team, hence this approach worked. When it came to integration time, where more teamwork would be required, our team mobilised reasonably well. We were able to get the work done which was the main thing, however we lacked the structure we had from the get-go. The work-balance became more uneven across members, which is something we should have probably worked on. As for my own contribution, one of my more collaborative roles was as the report editor. I had the task of getting everyone reading the particular sections they worked on, suggest edits, and from there improve the report. I think this iterative approach really helped the team as they were clued in more on what was going on.

2. The diversity of people really played in to our hands. We were able to utilise specific members' strengths in their particular field, whether it be programming or ADC. This allowed the team to make progress in lots of different areas simultaneously, which meant we had more time than others for integration. For example, whilst the electrical engineers designed their circuits, they didn't have to worry about the software, which I believe saved time.

3. I have found working on the project that, whilst it has built up some academic skills around problem solving, designing your own tasks, allowing a different perspective to a problem etc., I found that I mainly improved my soft skills. At times, I had to recognise that while my solution was correct, this may not be easy for someone else to work with. This was one of the reasons we digitised all our signals, so that software team could have an easier time managing the telemetry.

First Name: Ximan
Last Name: Wang
CID: 02374979
Group Number: 14

ELEC40006-Electronics Engineering Design Project 1

Reflection on Professional Practice Form

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All completed forms must be compiled and attached to the end of your group's project report.
Submit the full report, including all reflection forms, by 17:00pm on 11 June 2026.

1. How effectively did your group function as a team? How effective was your personal contribution to the team?
2. How did a diversity of people in the teams help you to perform better as a team?
3. What skills and knowledge did you acquire in this project? How?

1. Overall, our group functioned well together. We divided responsibilities early on according to each member's strengths and interests, which allowed different subsystems to progress in parallel without significant bottlenecks. Working together in the lab made communication straightforward — issues could be raised and resolved face to face, which meant subsystem integration went relatively smoothly throughout the project. My personal contribution centred on the IR sensing circuit design and the preparation of the interim presentation. For the IR circuit, I designed a signal conditioning stage to detect and distinguish IR pulses reflected from different rock surfaces, iterating through multiple circuit configurations and testing on breadboard. For the presentation, I coordinated input from all subsystems and compiled the slide deck, which required me to engage with each member's work beyond the scope of my own subsystem.

2. Having a team meant that we could support and encourage each other when facing difficulties rather than tackling problems alone. When someone got stuck, others would step in to discuss and brainstorm together, which often helped us look at problems from different angles and arrive at solutions that none of us might have reached individually. For me personally, this was especially valuable when working on the IR circuit. There were moments where I felt stuck and unsure how to proceed, and being able to talk through the problem with teammates helped me think more clearly and consider approaches I had not thought of on my own. Over time, this kind of exchange made me more confident in approaching unfamiliar problems and more open to reconsidering my initial assumptions when things did not go as expected.

3. This project gave me technical and professional skills that I had not fully developed through lectures alone. On the technical side, I was able to apply concepts such as transimpedance amplification, AC coupling, and comparator-based signal conditioning in a practical context, which deepened my understanding of them considerably. I also used LTspice for circuit simulation, which helped bridge the gap between theoretical knowledge and real circuit behaviour.

First Name: Colin
Last Name: Qian
CID: 02556839
Group Number: 14

ELEC40006-Electronics Engineering Design Project 1

Reflection on Professional Practice Form

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1. How effectively did your group function as a team? How effective was your personal contribution to the team?
2. How did a diversity of people in the teams help you to perform better as a team?
3. What skills and knowledge did you acquire in this project? How?

1. Our group functioned effectively because each member had a clear responsibility and was able to focus on a specific subsystem. This structure helped us avoid confusion and made progress easier to monitor. As team leader, I contributed by setting development guidelines, deadlines and priorities, while checking whether proposed plans were realistic for our available time, resources and technical ability. I also supported teammates during lab sessions, particularly when debugging or integrating different parts of the robot. My main contribution was keeping the project coordinated and ensuring that individual work combined into one complete system. Over time, the core team developed strong synergy because the team became more cohesive and trusted each other's roles.

2. The diversity of the team improved our performance because members had different technical strengths, working styles and approaches to problem-solving. This allowed us to divide tasks more effectively, such as control, PCB design, mechanical design and testing, according to each person's confidence and ability. It also meant that several ideas could be developed, discussed and compared before we committed to a solution. Different perspectives helped us identify weaknesses earlier, including possible issues in the IR circuit and the integration between hardware and software. When one member was stuck, others could approach the problem differently and help debug it faster.

3. I acquired both professional and technical skills during this project. Professionally, I learned that leadership requires continuous progress monitoring, clear communication, realistic timelines and the ability to stay calm under pressure. I also learned that planning should include backup solutions, since delays or failed designs are common in engineering projects. Managing the team taught me to assign tasks early, match responsibilities to each member's strengths and still give everyone meaningful involvement so they remained motivated. Technically, I learned to use FreeCAD and SolidWorks for 3D-printing design, KiCad for schematic and PCB development, Adafruit coding resources and datasheets for component selection. I also gained experience following a structured engineering process: planning, designing, implementing, testing and evaluating.